

SURVEY

Cognitive Architectures in Autonomous Robotics: A Systematic Review of Behavior Generation Approaches and Evaluation Strategies

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ABSTRACT Cognitive architectures are an essential component in the development of intelligent autonomous robots, enabling planning, learning, and adaptation to dynamic environments. This paper presents a systematic review of the literature focused on cognitive architectures applied to behavior generation in autonomous robotics. A structured search and selection process was conducted using multiple scientific databases, applying explicit inclusion, exclusion, and quality assessment criteria. From an initial set of 502 studies, 22 publications (2018–mid-2025) were selected for in-depth analysis. The review identifies a growing trend toward hybrid architectures that integrate symbolic reasoning with data-driven methods, such as neural networks and behavior trees. The types of architectures, tools used (such as symbolic planners, neural networks, and behavior trees), middleware employed (with a predominance of Robot Operating System), as well as the methods and metrics used in their evaluation, are analyzed. This review provides a detailed overview of the state of the art and highlights the technologies and approaches currently most widely used in the design of cognitive robotic systems.

INDEX TERMS Cognitive robotics, autonomous robots, systematic review.

I. INTRODUCTION

Autonomous robots have become a key technology capable of performing complex tasks in dynamic and unstructured environments without continuous human supervision. These systems rely on an array of sensors, including cameras, LiDAR, microphones, and proximity detectors, to perceive their surroundings and on actuators such as wheels, robotic arms, and specialized locomotion mechanisms to act upon them. Central to their ability to plan, learn, and adapt is the use of *cognitive architectures*, which provide the computational framework for decision-making, behavior generation, and continuous adaptation.

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Cognitive architectures in robotics aim to emulate key aspects of human cognition, including perception, memory, reasoning, and learning, to enable robots to go beyond reactive behavior and achieve goal-oriented autonomy. These architectures integrate symbolic reasoning techniques with data-driven methods (e.g., neural networks, behavior trees) to create flexible and adaptive control strategies. Within the broader field of cognitive robotics, they address fundamental challenges of knowledge representation, reasoning under uncertainty, and long-term autonomy [1].

Despite a large body of work in this domain, existing reviews have largely focused on taxonomic classifications or high-level overviews of cognitive systems rather than systematic analysis of implementation strategies and evaluation methodologies. Previous systematic reviews in this area include the work of [2] on cognitive architectures

for social robotics and [3] on general cognitive systems. The comprehensive survey by [3] provides an extensive taxonomy of cognitive architectures and categorizes behavior generation tools across various cognitive systems, offering valuable insights into the general landscape of cognitive architectures up to 2017. However, there remains a significant gap in systematic studies that specifically analyze the evaluation methodologies and implementation strategies used in cognitive architectures for behavior generation in autonomous robotics. Such systematic analysis is critical for assessing the effectiveness, reproducibility, and transferability of these architectures in real-world robotic settings, yet few studies have addressed this methodological dimension comprehensively.

This review operates under several key assumptions: (1) that cognitive architectures represent a distinct approach to autonomous robotics that goes beyond simple reactive control, (2) that behavior generation is a fundamental capability that can be systematically studied across different architectural approaches, and (3) that evaluation methodologies can be meaningfully compared despite the diversity of robotic platforms and application domains. The scope is limited to peer-reviewed publications from 2018 to mid-2025, focusing specifically on cognitive architectures that generate behaviors rather than those primarily concerned with perception or learning in isolation.

The main contributions of this paper are:

- A systematic review of cognitive architectures applied to autonomous robotics, with a specific focus on behavior generation approaches and their evaluation.
- A comprehensive mapping of selected studies (2018–mid-2025) to identify prevailing architectures, middleware platforms, such as the Robot Operating System (ROS), and evaluation methods.
- An analysis of emerging trends, such as the rise of hybrid architectures that integrate symbolic and sub-symbolic techniques, and the identification of research gaps and open challenges.

The remainder of this paper is organized as follows. Section II provides the necessary background on cognitive architectures and their role in autonomous robotics, presenting a comprehensive classification of architectural types (deliberative, reactive, and hybrid) and establishing the theoretical foundation for the review. Section III details the systematic review methodology, including the search strategy based on the PICOC (Population, Intervention, Comparison, Outcomes, Context) framework, inclusion and exclusion criteria, and quality assessment procedures applied to ensure methodological rigor. Section V presents the analysis of the 22 selected studies, systematically addressing each research question (RQ1–RQ4) with comprehensive data synthesis and visualization. Section VI provides an in-depth discussion of the findings, analyzing trends, identifying research gaps, and examining the implications. Finally, Section VII presents the conclusions of this study, highlighting the main contributions and outlining specific directions for future research.

II. BACKGROUND

In order to correctly analyze cognitive architectures, it is first necessary to identify the types of cognitive architectures that can be found. To this end, a classification of cognitive architectures is presented below. To carry out the classification of cognitive architectures, it has been decided to divide the classification of control architectures and the classification of cognitive architectures that can be found in the literature.

A. CLASSIFICATION OF CONTROL ARCHITECTURES

Following a classical approach, cognitive architectures can be considered control architectures. Following the study by [4], control architectures are divided into three types, which are presented in Figure 1.

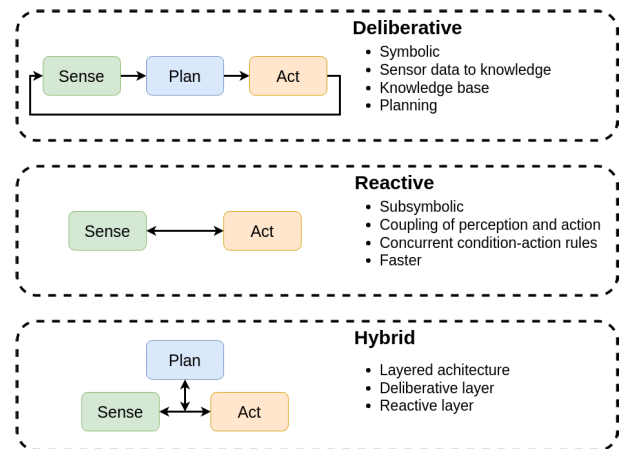


FIGURE 1. Classification of control architectures. There are three types of control architectures, which are as follows: deliberative architectures, which use symbolic knowledge and planners; reactive architectures, which use subsymbolic knowledge and have a high degree of coupling between sensors and actuators; and hybrid architectures, which combine the concepts of the previous two.

1) DELIBERATIVE ARCHITECTURES

The first type of control architecture is deliberative architecture. This type of architecture is based on the use of symbolic knowledge to define the robot's world. Symbolic knowledge is stored in a knowledge base that can be used for reasoning and planning processes. One of the most widespread ways in robotics to implement this type of architecture is through PDDL [5] which allows the robot's world to be defined and is compatible with multiple planners. Practical, high-performance planners that realize temporal and numeric planning (e.g., POPF) have been widely used in robotics and autonomous systems [6].

As presented in [7], these types of architectures are composed of three main phases: the perception phase, in which the robot obtains symbolic knowledge from the processing of sensory data; the planning phase, in which the robot generates plans or sequences of actions to solve specific objectives; and the action phase, in which the robot executes the actions that form the previously generated plan.

Various specialized tools can be used to carry out the planning and reasoning processes. For example, for planning, it is possible to use PDDL-based planners, such as POPF [6]. With regard to reasoning, ontologies can be used. Ontologies are knowledge structures that formally represent concepts, relationships, and properties within a specific domain, providing a solid framework for inference and reasoning in robot decision-making. These ontologies allow the robot to understand and process knowledge in a more profound and contextualized way, which is essential for informed decision-making and efficient interaction with its environment.

However, these architectures are slower than reactive architectures, as they require a planning step to generate the sequence of actions that will solve the robot's objectives. In addition, a preliminary step is required between the perception of the world and knowledge acquisition. This is known as the symbolic anchoring and symbol grounding problem [8], [9], which deals with the processes necessary to acquire symbolic knowledge from sensor data and to keep that knowledge linked to the real world. Probabilistic and graphical-model approaches to symbol grounding have been proposed to bridge low-level sensor measurements with high-level symbols in a principled fashion [10].

Recent research has also focused on integrating deliberative planners with learned models. Model-Based Reinforcement Learning (MBRL) [11] and Task and Motion Planning (TAMP) [12] techniques combine learned or approximate world models with symbolic planners to reduce sample complexity and improve robustness in complex, continuous domains. These efforts produce hybrid pipelines where planning is informed by learned dynamics and perception modules, thereby narrowing the gap between high-level symbolic planning and low-level sensorimotor control. The combination of learned world models with planners has proven particularly effective to improve sample efficiency and transfer in robotics, reducing the brittleness of purely deliberative or purely learned controllers [11].

2) REACTIVE ARCHITECTURES

The second type of control architecture is reactive architecture. This type of architecture is characterized by its ability to make quick and specific decisions based on real-time sensory information, making it suitable for tasks that require immediate and accurate responses.

By having greater coupling between sensor data and actuators, greater speed is achieved when generating behaviors. Thus, instead of using symbolic knowledge like deliberative architectures, reactive architectures use sub-symbolic knowledge, i.e., representations without direct meaning for humans, such as numerical values obtained from sensors.

Reactive architectures are based on a set of control modules, each of which specializes in a particular task. Each module takes data from the robot's sensors and generates commands for the actuators based on predefined rules or algorithms. This allows for efficient execution of specific

tasks and rapid response to changes in the environment. Often, these architectures are designed to operate in closed-loop control, where sensory information is constantly updated and decisions are made accordingly.

A common example of reactive architecture is the Subsumption Architecture [13], which is based on a hierarchy of simple, reactive behaviors that compete for control of the robot based on the relevance and urgency of situations. This approach has proven effective in mobile robotics applications such as autonomous navigation and obstacle avoidance.

While reactive architectures are ideal for tasks that require quick and specific responses, they often face limitations in their ability to perform complex tasks involving long-term planning or abstract reasoning. Furthermore, the use of sub-symbolic knowledge makes these architectures less transparent than deliberative ones.

Recent developments blur the line between reactive and learned systems: end-to-end deep policies and deep reinforcement learning approaches enable reactive-like low-latency behaviors while being trained from high-dimensional sensory inputs [14]. These emergent, learned controllers are increasingly used for real-time motor control and perception-to-action mappings, but require careful design to ensure safety, interpretability, and sample efficiency.

3) HYBRID ARCHITECTURES

The third type of control architecture is the hybrid architecture. These architectures represent an approach that seeks to combine the best of both worlds: reactivity and deliberation. They seek to overcome the inherent limitations of purely reactive or purely deliberative architectures, allowing autonomous robots to operate more efficiently and adaptively in complex and changing environments. An example would be AuRA [15].

Reactive architectures, with their focus on immediate decision-making based on sensory data, are ideal for tasks that require quick and specific responses. However, they lack the ability to plan for the long term and adapt to unforeseen situations. On the other hand, deliberative architectures are skilled at abstract reasoning and long-term planning, but can be inefficient in situations where immediate responses are needed.

Hybrid architectures seek to address these limitations by combining reactive and deliberative components into a single structure. Reactive modules enable rapid responses to sensory stimuli, while deliberative modules handle long-term planning and abstract reasoning. This combination offers the ability to adapt to changing situations while maintaining the ability to plan and reason. When organizing these modules, a layered approach such as that presented in [16] is often followed. This would result in a deliberative layer and a reactive layer.

Hybrid architectures have proven particularly effective in autonomous robotics applications, such as navigation in complex environments, object manipulation, and interaction

with humans. By balancing reactivity with deliberation, these architectures allow robots to adapt to a wide range of challenges in real time, making them a promising choice in the development of smarter and more versatile robots.

B. CLASSIFICATION OF COGNITIVE ARCHITECTURES

In literature reviews [3], [17], cognitive architectures are often classified into three categories similar to those of control architectures. This classification is presented in Figure 2.

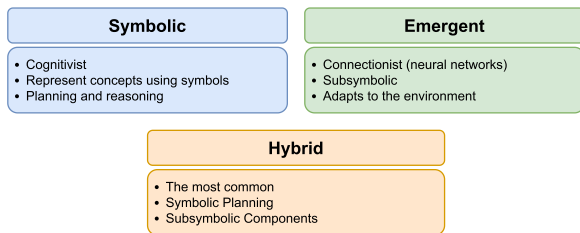


FIGURE 2. Classification of cognitive architectures. There are three types of cognitive architectures, which are as follows: symbolic architectures, which use symbolic knowledge, planning, and reasoning; emergent architectures, which use sub-symbolic knowledge and neural networks; and hybrid architectures, which combine the concepts of the previous two.

1) SYMBOLIC ARCHITECTURES

The first type of cognitive architecture is the symbolic architecture. These architectures are similar to the deliberative architectures. Both focus on information processing based on symbolic knowledge. Symbolic architectures are closely related to cognitivism, an approach in psychology and artificial intelligence based on the idea that the human mind processes information through symbolic representations and logical rules. Cognitivism holds that human thought can be understood through the manipulation of symbols, and these ideas have been applied in the creation of symbolic cognitive architectures in artificial intelligence, including those used in robotics.

Symbolic architectures are based on the idea that human cognition and reasoning can be modeled through symbolic representations and logical rules. These architectures use formal languages to represent information and rules that govern decision-making. They often operate with a hierarchical structure, where high-level goals can be set to guide task planning and execution.

In these architectures, robots can reason and make decisions based on symbolic information and logical rules. This makes them suitable for tasks that require a high level of abstraction and long-term planning, such as route planning in complex environments, natural language processing, logical inference, and decision-making in well-defined situations.

Examples of symbolic architectures include the Soar system [18], which is used in high-level applications such as decision-making in autonomous robotics and expert systems, and formal logic-based systems used for planning and problem-solving in robotics.

An important aspect of symbolic architectures is their contribution to safety, verification and explainability in robotic systems. As planners and learned components are increasingly combined in modern architectures, verification techniques for hybrid architectures and methods for explaining mixed symbolic-sub-symbolic decisions have become active research areas. The symbolic nature of these architectures makes them particularly suitable for formal verification and safety-aware planning, as seen in recent surveys on neuro-symbolic AI and safety-aware planning [11], [19].

2) EMERGENT ARCHITECTURES

The second type of cognitive architecture is the emergent architecture. These are not exactly the same as reactive architectures, although they share some similarities in their focus on perception-based decision-making and adaptation to changing environments. Emergent architectures are based on the idea that complex behaviors and capabilities can emerge from the interaction of multiple simple components, without the need for symbolic representations or logical rules.

Unlike reactive architectures, which are often based on a predefined set of rules and behaviors specific to particular situations, emergent architectures use neural networks and other machine learning techniques to generate behaviors in robots. In addition, they tend to have an enactive interpretation, meaning that these neural networks often learn and adapt from experience and feedback received from the environment, allowing them to adjust their behavior as they encounter new situations or data. However, this can undermine previously performed training.

Recent developments in emergent architectures have explored frameworks originating in computational neuroscience, such as predictive processing and active inference. These frameworks propose unifying views of perception, action and learning based on free-energy principles that are being explored as architectural principles for embodied agents [20], [21]. Additionally, continual and lifelong learning has become a critical challenge for emergent systems, as they face difficulties when adapting continually. Recent proposals explicitly target catastrophic forgetting, continual symbol acquisition, and online adaptation for long-lived agents [11], [19].

3) HYBRID ARCHITECTURES

The third type of cognitive architecture is the hybrid architecture. As in the case of the hybrid architectures in the previous classification, these architectures attempt to combine the best of two worlds: the symbolic and the emergent. They seek to overcome the problems of pure architectures, which are presented in Figure 3, allowing autonomous robots to access planning and reasoning capabilities as well as different emergent systems.

Symbolic architectures have a great capacity for planning and reasoning tasks and are highly explainable, but they require an initial state and have problems with changing

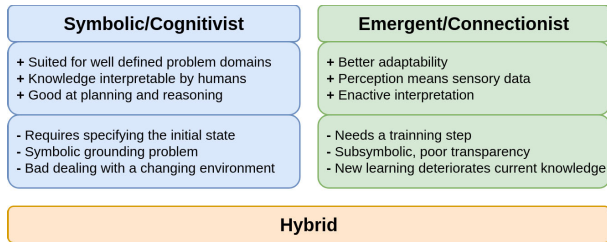


FIGURE 3. Advantages and disadvantages of cognitive architecture types. Symbolic architectures are well suited to problems with well-defined domains, their knowledge is interpretable by humans, and they are good at planning and reasoning; but they need to generate symbolic knowledge from sensors and an initial state, and they are poor in changing environments. Emergent architectures are better suited to environments, but since they require a training step and retraining implies worsening the models.

environments. On the other hand, emergent architectures are better suited to environments and do not require an intermediate step after obtaining data from sensors. However, they require a training phase, are not as transparent due to the use of sub-symbolic knowledge, and learning new things can deteriorate current knowledge.

As a result, hybrid architectures attempt to combine both components into a single structure. In this way, the resulting architectures can have a core formed by a symbolic planner that is responsible for long-term planning and several emergent components that allow actions to be performed with a greater ability to adapt to the robot's environment. Different types of hybridization can be found in the literature.

Recent advances in neuro-symbolic integration represent a significant trend in hybrid cognitive architectures. A growing body of work integrates symbolic reasoning and planners with neural perception and learning modules, with the explicit aim of combining interpretability and long-horizon planning with robust perceptual generalization [19]. This includes approaches that translate neural outputs to symbols and tightly coupled systems that interleave learning and reasoning. Furthermore, Large Language Models (LLMs) and multimodal Vision-Language Models (VLMs) are increasingly integrated as core components in robotic cognitive architectures. These foundation models can serve as high-level planners, communicators, and knowledge sources, generating action sequences or symbolic plans that are grounded via perception modules, affordance functions, or low-level skills [22], [23], [24], [25]. Agent-style architectures, including Voyager, Inner Monologue, and ReAct-inspired pipelines, leverage LLMs to dynamically decide when to invoke tools, low-level controllers, or motion primitives, enabling flexible hierarchies that combine deliberative reasoning with reactive execution [23], [26].

III. METHODOLOGY

To conduct this systematic review of the literature, we applied the PRISMA guide [27] and the methodology proposed by [28], which addresses the performance of this type of work in the field of software engineering. The general process required for conducting this type of review is presented in

Figure 4. The objective of this review is to obtain an overview of the current state of cognitive architectures used to generate behaviors in autonomous robots. Thus, the review focuses on identifying and analyzing the literature in journals that addresses the use of cognitive architectures in autonomous robots.

A. SEMIREFERENCE

As a first step in this literature review, we gathered a group of seven articles to form a semi-reference set. These articles were selected based on our preliminary search, the authors' existing familiarity with the field, and recommendations from other experts. This set served to identify key keywords and shape the research direction. Furthermore, it will act as a validation mechanism for our full search methodology, which is elaborated upon in III-C3. The complete list of papers in the semi-reference set is provided in Table 1.

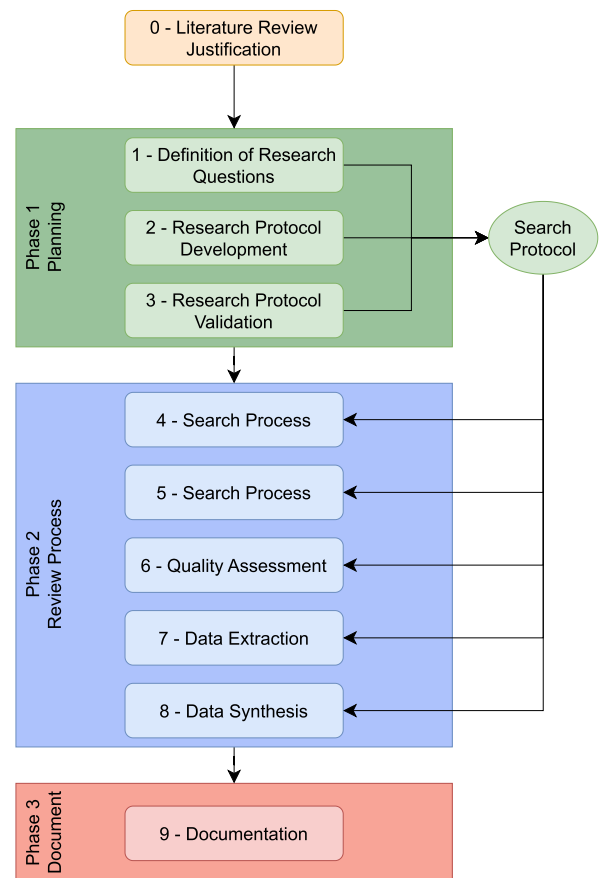


FIGURE 4. General structure of a systematic review process. Figure adapted from [28].

B. RESEARCH QUESTIONS

The first step in conducting a systematic review is to define the research questions. The PICOC strategy [35] provides a fundamental structure for formulating these questions. This methodology evolves from the traditional PICO strategy, as presented by [36] in 2007. Thus, the PICOC strategy

TABLE 1. Semi-reference set of papers used to guide the systematic review methodology and validate the search strategy.

#	Reference
SR1	[3]
SR2	[29]
SR3	[30]
SR4	[31]
SR5	[32]
SR6	[33]
SR7	[34]

focuses on identifying the following key elements in a research paper:

- **Population:** Who are the subjects of the study or what is the problem being investigated?
- **Intervention:** What intervention, treatment, or exposure is being examined?
- **Comparison:** Is there a comparison group or alternative to the intervention being evaluated?
- **Results:** What are the expected results or measures that will be used to evaluate the intervention?
- **Context:** What is the context in which the research is being conducted?

In this research project, the population is cognitive architectures and the context is robotics. The intervention corresponds to the tool and components used to generate behaviors in robots. The desired results are the current cognitive architectures and the tools used to generate behaviors in autonomous robots. No specific comparisons are sought.

Based on the concepts identified above, the following research questions are constructed:

- **RQ1:** What cognitive architectures does the state of the art consider for autonomous robots?
The first research question focuses on understanding the state of the art in generating autonomous behaviors in robots through cognitive architectures. The objective of this question is to find out what architectures currently exist and what characteristics they have.
- **RQ2:** What tools are used in cognitive architectures to generate behaviors in autonomous robots?
The second research question focuses on identifying the tool used by cognitive architectures to generate behaviors in robots. The objective of this question is to find out which tools are currently used by cognitive architectures in order to assess their inclusion in the cognitive architecture to be developed in this research project.

It has been decided to include secondary questions related to the evaluation of cognitive architectures. Since this is not the main objective of this work, these questions have not been included with the previous ones. However, given the highly diverse and complex nature of cognitive architectures in robotics, as well as the variations in the robotic platforms used, the inclusion of these questions allows these particular

differences and challenges to be addressed in a systematic manner. Thus, the following questions are posed:

- **RQ3:** What methods are used to evaluate cognitive architectures?
The third research question focuses on understanding the methods and experiments used to evaluate cognitive architectures. Since architectures are used on different robotic platforms and for different purposes, their evaluation will differ. The objective of this question is to understand what evaluation methods exist in current cognitive architectures.
- **RQ4:** What metrics are used to evaluate cognitive architectures?
The fourth research question focuses on identifying the metrics used in the evaluation of cognitive architectures. The objective of this question is to identify the metrics used in current architectures during the experimentation and evaluation phases.

C. SEARCH PLANNING

The search process was designed by first determining the relevant information sources, applying necessary restrictions, and defining a strategy to validate the results.

1) INFORMATION SOURCES

The selection of information sources was guided by the need to ensure comprehensive coverage of the cognitive robotics literature while maintaining focus on high-quality, peer-reviewed publications. Six major scientific databases were selected based on their coverage of robotics, artificial intelligence, and computer science literature:

- **ACM Digital Library:** Provides comprehensive coverage of computer science publications, including robotics and AI conferences and journals.
- **IEEE Xplore Digital Library:** Essential for robotics research, covering IEEE Robotics and Automation Society publications and related conferences.
- **Web of Science:** Multidisciplinary database providing broad coverage across science and technology domains with strong citation indexing.
- **ScienceDirect:** Elsevier's platform covering engineering, computer science, and technology journals relevant to robotics research.
- **Scopus:** Comprehensive abstract and citation database covering peer-reviewed literature in science, technology, and engineering.
- **Springer Link:** Major publisher platform covering computer science, engineering, and robotics publications.

2) SEARCH LIMITATIONS

Several limitations were identified and addressed in the search strategy:

- **Language Restriction:** The search was limited to English-language publications to ensure consistency in

analysis and due to resource constraints for translation and review of non-English papers.

- **Temporal Scope:** Publications were restricted to the period 2018-mid 2025 to focus on recent developments while maintaining a manageable corpus size. This may exclude some foundational work, but it ensures relevance to current research trends.
- **Publication Type:** The search focused on peer-reviewed journal articles and papers from top-tier robotics conferences (IROS, ICRA, HRI, AAMAS) to ensure quality and impact, potentially excluding relevant work from workshops or technical reports.
- **Search Field Limitation:** Searches were conducted on the title, abstract, and keywords fields only, which may miss relevant papers where key concepts appear only in the full text.
- **Terminology Variations:** The search string was designed to capture standard terminology, but variations in how cognitive architectures are described across different research communities may result in some relevant papers being missed.

3) SEARCH PROCESS VALIDATION

After executing the search, we collected the returned papers. Validating this initial set is crucial for confirming that the search was performed correctly. We employed the validation criterion from [37], which mandates that at least 70% of the semi-reference set appear within the search results. If this condition was satisfied, the search was considered valid. If not, we iterated by reformulating the search string to enhance the result set's accuracy.

D. SEARCH PROCESS

The search for information is carried out automatically using various renowned digital databases, namely the ACM Digital Library, IEEE Xplore Digital Library, Web of Science, ScienceDirect, Scopus, and Springer Link. The main objective of this search is to identify and compile papers published up to July 13, 2025, that are related to the field of interest, which is the use of cognitive architectures to generate behaviors in autonomous robots. Special emphasis is placed on examining the title, abstract, and keyword fields of the papers, and those written in languages other than English are deliberately excluded in order to maintain consistency and uniformity in the analysis of the information.

In addition, it is important to note that an additional process is carried out to construct a precise and specific search string. This is done by extracting relevant keywords, which are obtained from the research questions posed in the study. For this purpose, a previously defined structure known as PICOC was used, which encompasses the essential elements of the research. These PICOC fields are consistent with those mentioned above. However, in the field of intervention, the use of middleware was incorporated, in addition to tools for generating robot behavior. Among the most prominent

middleware in the field of robotics are ROS (Robot Operating System) [38], ROS 2 (Robot Operating System 2) [39], YARP (Yet Another Robot Platform) [40], NAOqi [41], and RoboComp [42]. This is particularly important, as cognitive architectures in robotics require middleware to establish communication with the robotic platform on which they will be implemented.

The inclusion of middleware in the search criteria is justified for several reasons. First, cognitive architectures cannot operate in isolation and must interface with physical robotic systems through standardized communication protocols provided by middleware platforms. Second, the choice of middleware significantly influences the architectural design patterns, available tools, and integration capabilities of cognitive systems. Third, middleware platforms often provide essential services such as message passing, service discovery, parameter management, and hardware abstraction that are fundamental to cognitive architecture implementation. Finally, by including middleware terms in the search string, we ensure the identification of studies that demonstrate practical implementation of cognitive architectures rather than purely theoretical or simulation-based approaches, thereby focusing on systems that have been validated in realistic robotic environments.

Consequently, a PICOC has been defined that encompasses the following key elements:

- **Population:** Cognitive Architectures
- **Intervention:** Tools for generating behaviors and Middleware (ROS, YARP, NAOqi, RoboComp)
- **Comparison:** Not applicable in this context
- **Outcome:** Identification of current cognitive architectures and tools used to generate behaviors in autonomous robots
- **Context:** Robotics

Using these PICOC components as a starting point, the essential keywords that make up the resulting search string are generated. This search string, which is used in all information sources, is presented below:

("Cognitive Architecture") AND ("Robotics") AND ("ROS" OR "Robot Operating System") OR ("YARP" OR "Yet Another Robot Platform") OR "RoboComp" OR "NAOqi")

E. SELECTION PROCESS

The complete search and selection process is shown in Figure 5. The first stage consists of searching for papers using the previously defined string. The second stage of the process involves removing duplicates from the search results. The selection process was then designed around three phases: filtering by title, filtering by abstract, and filtering by full text. In each of these stages, the corresponding section of the paper or the full text was reviewed and the inclusion and exclusion criteria defined below were applied. This allowed for a gradual filtering of papers that were not relevant to the research topics proposed.

To carry out the paper selection and filtering process, a set of inclusion criteria and a set of exclusion criteria have been defined, which are detailed in Tables 2 and 3, respectively. For a paper to pass the process, it must meet all the inclusion criteria and not meet any of the exclusion criteria.

F. QUALITY ASSESSMENT PROCESS

Once the planning stage of the selection process has been completed, a quality assessment of the papers is carried out. In this phase, each of the previously selected papers is assigned a score, establishing a minimum threshold that they must exceed in order to be considered for review.

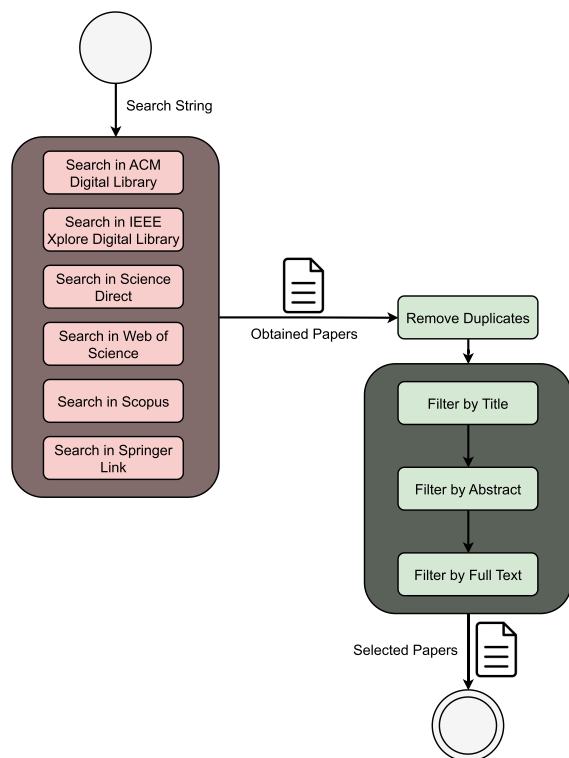


FIGURE 5. Diagram of the search and selection processes. The search process, marked in red, uses the search string to search for papers in the ACM Digital Library, IEEE Xplore Digital Library, Web of Science, ScienceDirect, Scopus, and Springer Link. The selection process, marked in green, has been divided into duplicate removal and several filtering processes: by title, by abstract, and by full text.

The quality assessment is carried out by answering a series of questions related to each of the papers previously selected in the previous stage. For this purpose, the set of questions proposed by [43] and cited by [28] in the reference guide used for the preparation of this review has been adopted. The questions have been modified, removing those that were not relevant to this review and adding new ones. In this way, the final set of questions has been formed, which is detailed in Table 4. These questions cover the four main areas of a research paper:

TABLE 2. Inclusion criteria for the selection process.

#	Description
IC1	Research paper from a journal or a top robotics conference: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), The International Conference on Robotics and Automation (ICRA), ACM/IEEE International Conference on Human-Robot Interaction (HRI), and Autonomous Agents and Multiagent Systems (AAMAS)
IC2	Entire paper available in English.
IC3	All or part of the paper's content must directly answer one of the research questions.
IC4	Publication date after 2017 (not included).

TABLE 3. Exclusion criteria for the selection process.

#	Description
EC1	Papers not written in English.
EC2	Papers not related to robotics or cognitive architectures.
EC3	Papers that do not present a cognitive architecture.

- **Communication:** These questions focus on determining the quality of the presentation and communication of the paper.
- **Rigor:** This includes questions related to the design of the research carried out in the paper.
- **Credibility:** These questions are intended to assess whether the research results are valid and meaningful.
- **Relevance:** These questions focus on determining the relevance of the information presented in the paper.

TABLE 4. Questions from the paper quality assessment. They have been classified into four areas: communication, rigor, credibility, and relevance.

Area	Question
Communication	Is there a clear definition of the research objectives?
Communication	Is there a clear definition of the conclusions?
Communication	Does it contain an adequate description of the research context?
Communication	Is there a clear diagram of the cognitive architecture?
Communication	Are the components of the cognitive architecture described?
Rigor	Is the research design appropriate?
Rigor	Was a validation or experimentation phase carried out during the research?
Credibility	Has the cognitive architecture been tested on real robots?
Credibility	Is there a clear reference to the results obtained?
Relevance	Is the study valuable for research or practice?

Each of the above questions offers four possible answer options: “Yes,” “Partially,” “No,” and “Not applicable.” These are assigned scores of 1, 0.5, and 0, respectively, with “Not applicable” excluded from the total possible score. A quality threshold of 7.5/10 was established for inclusion in the review. This threshold was selected over a lower one (e.g., 5.0, or half the maximum score) to ensure a high

standard of methodological rigor and relevance. A score of 7.5 requires a paper to consistently achieve “Yes” or “Partially” across the majority of its applicable criteria, ensuring it demonstrates substantial quality in multiple key aspects. This prevents the inclusion of papers that may answer generic questions correctly but fail on more specific, critical points of evaluation.

To assess the robustness of this threshold, alternative thresholds of 7.0 and 8.0 were considered during the methodology design phase. A threshold of 7.0 was deemed potentially too permissive, as it might allow inclusion of studies with significant methodological limitations across multiple quality criteria. A threshold of 8.0 was considered overly restrictive, potentially excluding relevant studies that demonstrate high quality in most areas but have minor limitations in specific aspects. The 7.5 threshold was selected as it optimally balances inclusivity with methodological rigor, ensuring that included studies consistently demonstrate substantial quality across the evaluation criteria while maintaining sufficient breadth to capture diverse approaches in cognitive architectures for autonomous robotics.

G. DATA EXTRACTION AND SYNTHESIS PROCESSES

The data extraction stage aims to extract data directly related to the review and research questions. For example, cognitive architectures, middlewares, tools for generating behaviors, and metrics and evaluation methods. Thus, the following key questions have been defined to extract information:

- What middleware is used in the paper?
- What type of cognitive architecture is presented in the paper?
- Is it a layered architecture?
- What behavior generation tools are used?
- How is the cognitive architecture evaluated?
- What metrics are used to evaluate the cognitive architecture?

Based on the extracted data, a synthesis of this information will be carried out in the form of tables, where both the reference and a unique identifier for each paper will be recorded. In addition, the data collected for each of the papers will be established based on the research questions they answer.

IV. RESULTS

The numerical results of this systematic review are presented in the PRISMA (*Preferred Reporting Items for Systematic Reviews and Meta-Analysis*) diagram in Figure 6. The results obtained in the search, selection, quality assessment, and data synthesis phases are explained in more detail below.

A. SEARCH PROCESSES

The processes of searching for and selecting papers for the review process are summarized in the PRISMA diagram shown in Figure 6. A total of 502 papers were found in the paper search process, divided among the different search sources as detailed below:

- ACM Digital Library: 47 papers
- IEEE Xplore Digital Library: 9 papers
- ScienceDirect: 12 papers
- Web of Science: 80 papers
- Scopus: 15 papers
- Springer Link: 339 papers

We applied the validation method to the resulting set. The search process was considered valid since the set contains 6 of the 7 articles from the reference set, which corresponds to approximately 86%.

B. SELECTION PROCESSES

The selection process began with the elimination of duplicate papers, 38 in total, leaving the number of candidate papers at 464. Subsequently, the papers were filtered by applying the inclusion and exclusion criteria, obtaining a total of 37 candidate papers and excluding 428 papers.

C. QUALITY ASSESSMENT PROCESS

After searching for and selecting papers, a quality assessment was carried out. Table 5 summarizes the results of the quality assessment carried out on the 37 selected papers.

TABLE 5. Quality assessment of the selected papers, showing their reference, identifier, and score.

Reference	Identifier	Score
[44]	S1	8.0
[45]	S2	3.0
[46]	S3	8.0
[47]	S4	5.0
[29]	S5	9.0
[48]	S6	5.0
[32]	S7	9.0
[49]	S8	6.5
[31]	S9	9.0
[50]	S10	8.0
[51]	S11	7.5
[52]	S12	8.0
[53]	S13	7.0
[30]	S14	9.5
[54]	S15	8.0
[55]	S16	7.5
[56]	S17	5.0
[57]	S18	8.5
[33]	S19	9.0
[34]	S20	9.5
[58]	S21	7.5
[59]	S22	8.5
[60]	S23	7.0
[61]	S24	9.5
[62]	S25	9.5
[63]	S26	8.5
[64]	S27	7.0
[65]	S28	9.0
[66]	S29	6.0
[67]	S30	8.5
[68]	S31	9.0
[69]	S32	9.0
[70]	S33	7.5
[71]	S34	7.0
[72]	S35	8.0
[73]	S36	8.5
[74]	S37	7.0

Of the 37 papers previously selected, 22 pass the quality assessment process. Fifteen papers are excluded. Table 6 shows the 22 final papers, which are assigned an identifier for reference later on.

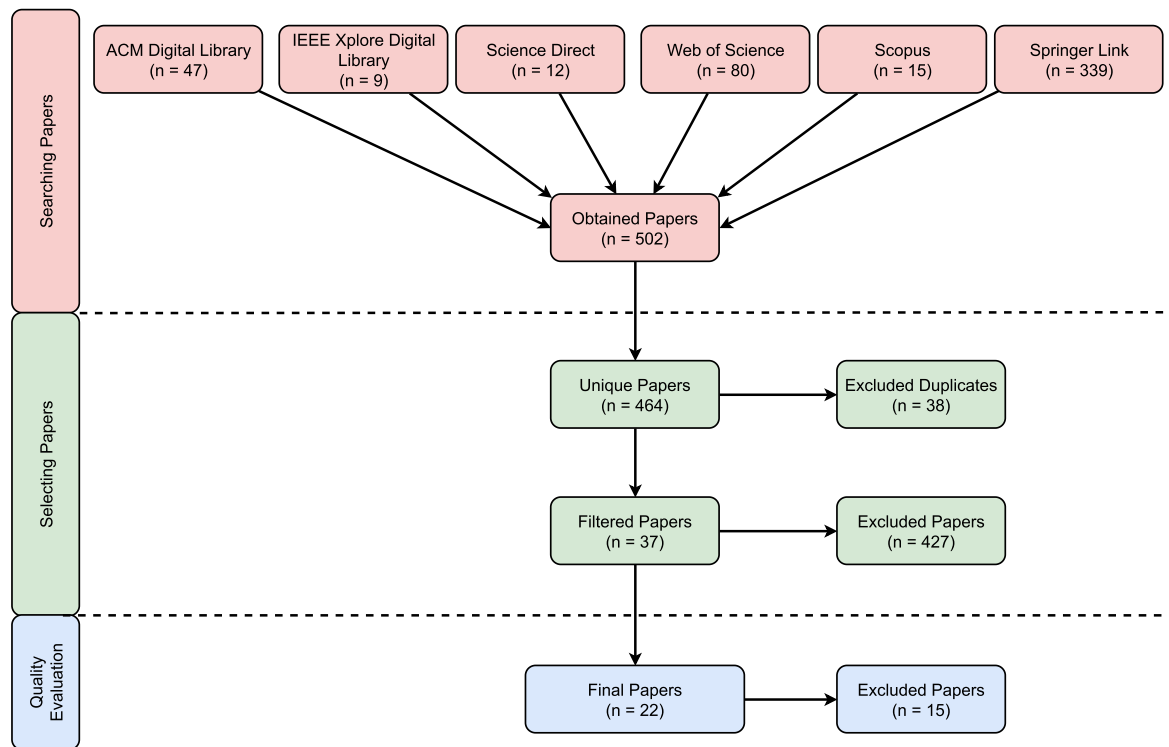


FIGURE 6. PRISMA diagram of the paper search, paper selection, and quality assessment processes. Initially, 502 papers were found in the search (47 papers from ACM Digital Library, 9 papers from IEEE Xplore Digital Library, 12 papers from ScienceDirect, 80 papers from Web of Science, 15 papers from Scopus, and 339 papers from Springer Link). After selection, 37 papers were obtained. Finally, after quality assessment, 22 papers were obtained.

TABLE 6. Final papers that have passed the evaluation process. Each paper has a new identifier and reference.

ID	Reference
FS1	[44]
FS2	[46]
FS3	[29]
FS4	[32]
FS5	[31]
FS6	[50]
FS7	[52]
FS8	[30]
FS9	[54]
FS10	[57]
FS11	[33]
FS12	[34]
FS13	[59]
FS14	[61]
FS15	[62]
FS16	[63]
FS17	[65]
FS18	[67]
FS19	[68]
FS20	[69]
FS21	[72]
FS22	[73]

D. DATA EXTRACTION AND SYNTHESIS

In the data extraction and synthesis stage, data related to the questions presented above were obtained. Thus, Tables 7, 8, 9, and 10 present the information extracted after reading the papers that were selected following the quality assessment.

V. DATA ANALYSIS

This section presents the analysis of the results. To this end, the research questions are answered using the data obtained during the review.

A. RQ1-COGNITIVE ARCHITECTURES IN AUTONOMOUS ROBOTICS

As presented in the classification of cognitive architectures, three main types are identified: symbolic (or deliberative) architectures, which use planners and deliberative systems for decision-making; emergent (or reactive) architectures, which are mainly based on neural networks and reactive behaviors; and, finally, hybrid architectures, which integrate elements of both approaches.

From the analysis of the 22 papers reviewed, presented in Table 7, it is evident that hybrid architectures are the most widely used, being present in 13 of the 22 studies. This is followed by reactive/emergent architectures, which appear in 5 papers, and symbolic/deliberative architectures, present in 4 papers. This predominance of hybrid architectures, reflected in Figure 7, suggests a clear trend toward solutions that balance planning and adaptability, suitable for dynamic environments and complex tasks.

Regarding layered organization, as shown in Figure 8, only 8 papers explicitly mention the use of a layered structure. Although this number is not a majority, it is relevant to

TABLE 7. Data extracted from papers accepted for the first three questions of the data extraction. For each paper, its identifier, the middleware it uses, the type of architecture, and whether a layered organization is used are shown.

ID	Middleware	Type of Architecture	Layers	Year
FS1	ROS	reactive/emergent	Yes	2018
FS2	ROS	hybrid	No	2018
FS3	ROS	hybrid	Yes	2018
FS4	ROS	hybrid	Yes	2019
FS5	RoboComp	hybrid	No	2019
FS6	ROS	hybrid	No	2019
FS7	ROS	reactive/emergent	No	2020
FS8	ROS	hybrid	Yes	2020
FS9	ROS	hybrid	No	2021
FS10	ROS	hybrid	Yes	2022
FS11	ROS 2	hybrid	Yes	2022
FS12	ROS	hybrid	No	2023
FS13	ROS	deliberative/symbolic	No	2023
FS14	ROS	reactive/emergent	No	2023
FS15	ROS	deliberative/symbolic	No	2023
FS16	ROS	reactive/emergent	No	2024
FS17	ROS 2	hybrid	Yes	2024
FS18	ROS	deliberative/symbolic	No	2024
FS19	YARP	reactive/emergent	No	2025
FS20	ROS	deliberative/symbolic	No	2025
FS21	ROS	hybrid	Yes	2025
FS22	RoboComp/ROS 2	hybrid	No	2025

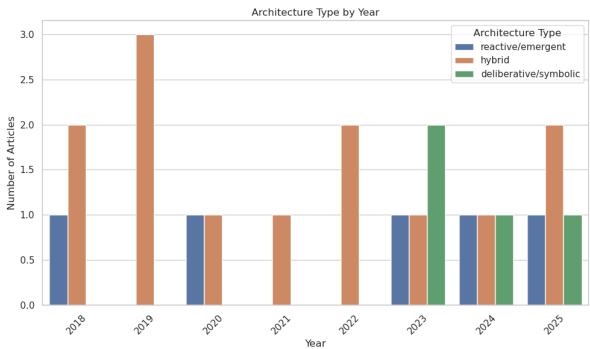


FIGURE 7. Types of architectures by year of the analyzed papers.

note that several papers that do not explicitly indicate it present a functional structure that clearly resembles a layered organization, which can be interpreted as an implicit implementation of this strategy.

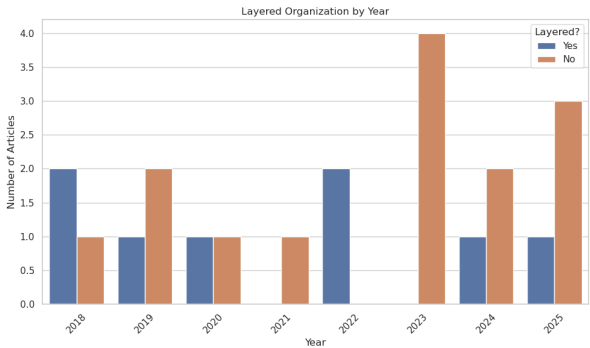


FIGURE 8. Layered architectures by year of the papers analyzed.

In terms of the middleware used, ROS is identified as the most widespread, present in 17 of the 22 papers, as shown in Figure 9. It is followed by ROS 2, used in three studies, and

other middleware such as RoboComp (two papers) and YARP (one paper). It is worth mentioning that one study (FS21) combines both ROS 2 and RoboComp, reflecting a mixed or transitional approach between platforms. This predominance of ROS and ROS 2 indicates their role as the de facto standard in the development of cognitive robotic systems, although it should be noted that the last distribution of ROS, which is Noetic, reached its EOL at the end of May 2025 [75], so ROS 2 would be a better alternative for future developments.

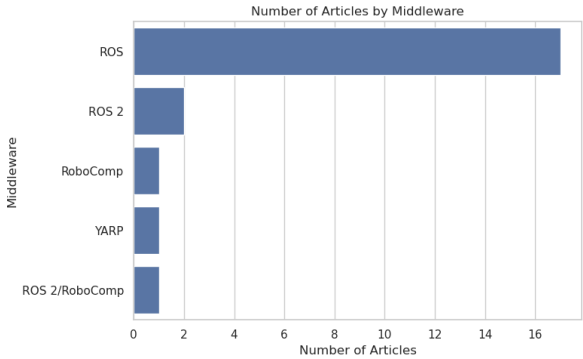


FIGURE 9. Robotics middleware by year.

If we look at the temporal distribution of architectures, we can see a gradual increase in the use of hybrid architectures in recent years (2023–2025), suggesting a consolidation of this approach in the current literature. Symbolic architectures reappear occasionally in 2023 and 2024, and emerging architectures remain present from 2018 to 2025, although to a lesser extent. Likewise, a slight transition from the use of ROS to ROS 2 is detected in the most recent works, which could be motivated by the improvements that this version offers in terms of communication, performance, and compatibility with distributed systems.

At last, the application domain breakdown reveals distinct patterns across different robotic applications. Service robots (FS2, FS6, FS11, FS17) demonstrate a consistent preference for hybrid architectures throughout the study period, reflecting the need to balance reactive responsiveness for human interaction with deliberative planning for complex service tasks. Human-robot interaction studies (FS1, FS10, FS15, FS19) show evolution from purely reactive approaches in early years (2018) toward hybrid architectures in recent studies (2022-2025), indicating increasing sophistication in social cognition requirements. Manufacturing and industrial applications (FS8, FS13, FS18, FS20) exhibit a notable shift toward deliberative architectures in recent years (2023-2025), suggesting that controlled environments increasingly benefit from explicit symbolic reasoning for complex task planning and safety verification. Exploration and navigation tasks (FS3, FS4, FS7, FS9) maintain consistent use of hybrid architectures across all years, reflecting the fundamental need to combine reactive obstacle avoidance with deliberative path planning in dynamic environments.

TABLE 8. Data extracted from papers accepted for the question on tools used to generate behaviors. For each paper, its identifier is shown along with a paragraph summarizing the tools used to generate behaviors.

ID	Tools for Generating Behaviors
FS1	This collaborative cognitive architecture is mainly based on teleoperation and navigation in robots. It also uses a state machine to control when to switch from user teleoperation to autonomous navigation.
FS2	This architecture uses a state machine to control its operation. This state machine is responsible for calling the planner, which is based on PDDL, and executing actions. It also contains an object detection and recognition system based on the use of point clouds and histograms. Finally, it uses a memory system, one to store object detection data and another to store plans, action results, and information about the robot's world.
FS3	HiMoP architecture is based on the use of PDDL-based planning together with state machines to generate behaviors. In addition, it features a motivational system that guides the robot's goals. Finally, it uses navigation and speech recognition.
FS4	MoBAR is an architecture that uses PDDL-based symbolic planning [5] and state machines.
FS5	CORTEX is a cognitive architecture based on the use of a graph that maintains symbolic knowledge and geometric information about the robot. The architecture has a set of agents that use this graph. Some of these agents perform the following tasks: localization, navigation, dialogue, manipulation, object detection, and speech recognition and synthesis. In addition, it is possible to include other tools such as PDDL-based planning, state machines, and behavior trees.
FS6	This architecture uses symbolic representations of the actions that the robot can perform. In addition, a neural network is used to control the activation of the robot's actions. The robot has a memory in which the robot's actions and their dependencies are defined symbolically. The neural network uses this memory.
FS7	This architecture uses an attention system to focus on certain people, such as the person who is speaking at the moment.
FS8	This layered architecture uses a deliberative system based on symbolic knowledge written in PDDL and symbolic planning using ROSPlan [76], and state machines created with SMACH [77]. These state machines are used to control the operation of the architecture and implement robot actions that use navigation, object recognition, speech recognition, and speech synthesis systems.
FS9	This cognitive architecture is based on the ACT-R model by [78]. In this way, it combines symbolic knowledge and planning with other components such as speech recognition, speech synthesis, and MoveIt [79] to control the robot's manipulators.
FS10	This hybrid architecture uses symbolic knowledge based on PDDL and is organized as a knowledge graph to define the state of the robot's world. Symbolic knowledge is used alongside ROSPlan [76], a planning system, to generate robot actions that are implemented as state machines. These state machines are used in several layers of the architecture.
FS11	This cognitive architecture uses a deliberative system based on symbolic knowledge written in PDDL and PlanSys [80], together with behavior trees to control the robot's behaviors.
FS12	This architecture has a symbolic planning and reasoning system and a reactive system. Planning and reasoning are based on the use of ontologies. On the other hand, the reactive system contains the robot's actions and performs natural language processing, speech synthesis, and speech recognition. Finally, both systems are communicated through Hierarchical Reinforcement Learning (HRL) [81].
FS13	This deliberative architecture uses symbolic knowledge in PDDL and ontologies. In this way, it defines the state of the robot's world symbolically. In addition, it uses ROSPlan, a PDDL-based planning system, to generate the robot's behaviors.
FS14	This architecture is based on the use of different emerging components. It uses different types of neural networks to carry out its operations. In this way, it uses convolutional neural networks (CNN) to recognize human user actions from images obtained from a camera. It also uses Long Short-Term Memory (LSTM) networks to predict how long it will take the user to perform an action. Finally, it contains a memory system to store predefined user data and actions, to communicate between the different components, and to store episodic information that can be analyzed later.
FS15	This architecture relies heavily on symbolic methods, including semantic knowledge and ontologies for contextual reasoning, and a hierarchical task network (HTN)-like structure for symbolic planning and task decomposition. Semantic graphs are used to map verbal cues to relevant tasks, while an activation spreading mechanism governs task progression through nodes with defined states. Object detection is performed using ROS-based AR tags, and user interaction is facilitated through speech-to-text processing and likely text-to-speech responses, with intent management handled by a Rasa chatbot [82]. Finally, MoveIt is used to control the robot's arms.
FS16	This cognitive architecture combines a motivational system with internal motivations and external stimuli through competitive selection, modulated by emotional states. They integrate multimodal perception (vision, touch, sound) and make decisions based on homeostasis, choosing between appetitive (search) and consummatory (satisfaction) actions to adapt autonomously and coherently to the environment.
FS17	This layered architecture uses its own deliberative system, based on symbolic knowledge written in PDDL [5], and behavior trees and state machines created with YASMIN [83]. These state machines are used to control the operation of the architecture and implement robot actions that use navigation, object recognition, speech recognition, and speech synthesis systems. Finally, it also presents a minor inclusion of explainability.
FS18	This architecture, DIARC, is based mainly on the use of symbolic knowledge, written in PDDL [5], which is obtained from the robot's sensors. In addition, there is a planning system to manage the robot's decision-making.
FS19	The architecture proposed in this work uses a state machine to control its components. In addition, it has a set of emerging modules for sound detection, speech recognition, speech synthesis, and text generation using Large Language Models (LLMs) [84, 85].
FS20	This architecture uses symbolic knowledge and AND/OR graph networks to carry out planning and decision-making.
FS21	The presented architecture makes use of elements of symbolic and deliberative architectures. In this way, it has a symbolic memory based on graphs. This memory is converted into PDDL [5], which will later be used by ROSPlan [76] to carry out planning. Finally, the architecture consists of object and face recognition, speech recognition, and speech synthesis.
FS22	The architecture of this work makes use of symbolic knowledge based on a knowledge graph and PDDL [5] and planners. In addition, it uses behavior trees to define the robot's actions. Finally, the architecture integrates object and speech recognition systems and speech synthesis to interact with its environment.

B. RQ2-TOOLS FOR BEHAVIOR GENERATION IN COGNITIVE ARCHITECTURES

Based on the analysis of the cognitive architectures described in the selected papers, various tools used to generate behaviors in autonomous robots are identified, which are presented in Table 8. Likewise, the tools selected within the analyzed papers are presented in Figure 10, which are symbolic planning, symbolic knowledge, knowledge graph, ontologies, state machine (FSM), behaviour trees (BTs), neural networks, object recognition, speech-to-text (STT), text-to-speech (TTS), reinforcement learning (RL), motivational system, and attention system.

As can be seen in Figure 11, the most frequently used tools are those related to deliberative systems, such as symbolic planning systems and symbolic knowledge. Next are reactive systems, such as state machines and emerging modules (object recognition, speech recognition, and speech synthesis), which are the most prevalent elements in the analyzed works. The least frequently used tools are ontologies, neural networks, motivational systems, reinforcement learning, and attention systems.

Finally, the co-occurrence of tools in the papers was analyzed. To do this, we used the heatmap shown in Figure 12. Each value indicates the proportion of co-occurrence relative

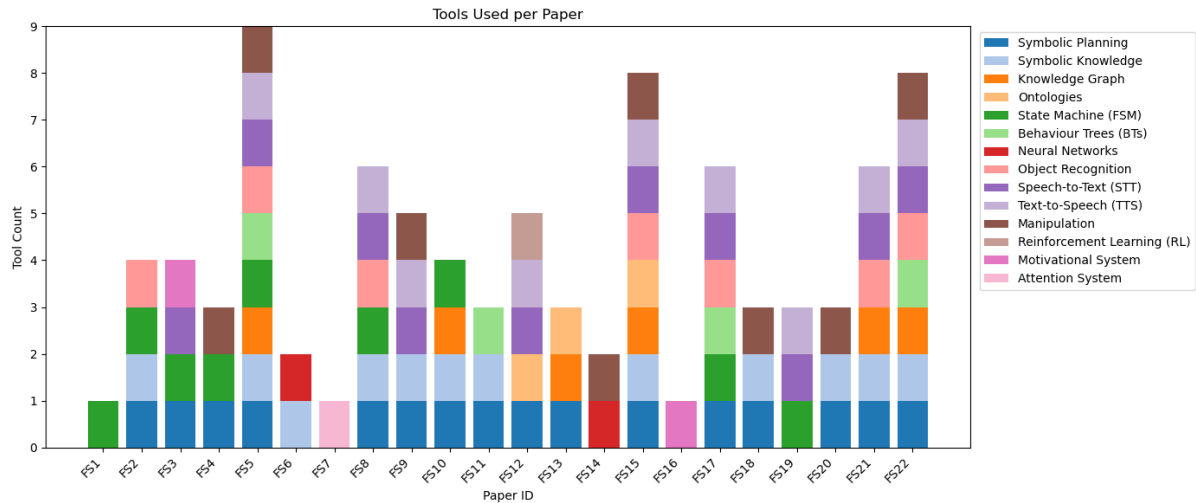


FIGURE 10. Graphical representation of the distribution of tools used in each paper analyzed.

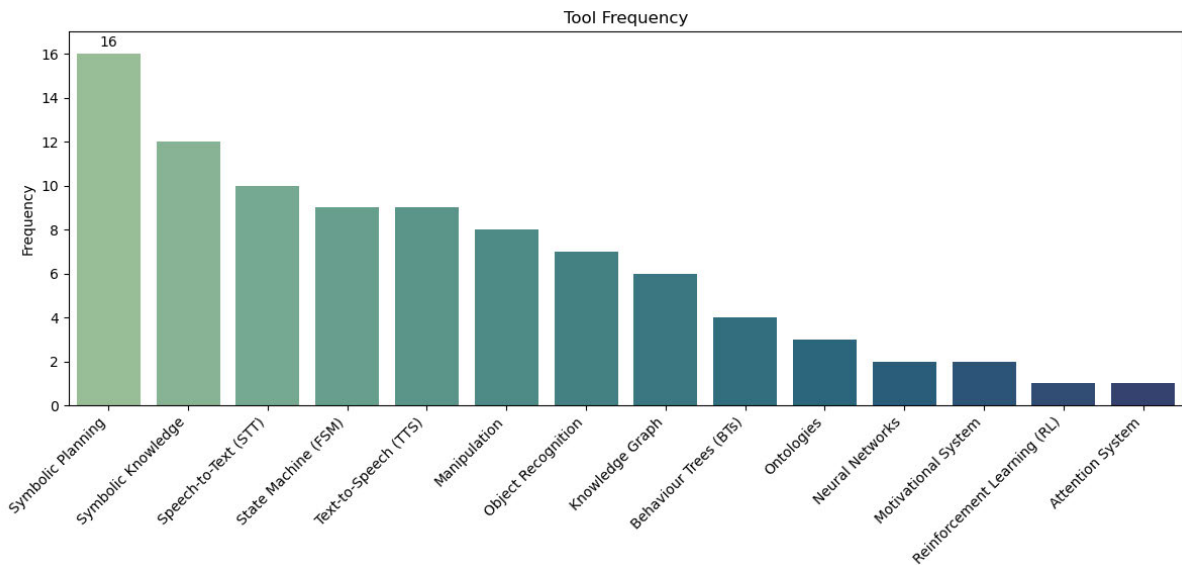


FIGURE 11. Frequency of occurrence of the tools used in the papers analyzed.

to the maximum absolute value of the entire matrix. Since normalization was performed by dividing each cell by the global maximum value, cells with a value of 1.0 represent the most frequent combinations of tools in the entire set of papers, while values close to 0 indicate very low or non-existent co-occurrence. There is a high co-occurrence between symbolic planning, symbolic knowledge, knowledge graphs, state machines, and object recognition, suggesting that these tools form a common basis in many architectures. Manipulation also presents a high co-occurrence with symbolic systems. Likewise, there is a strong association between interaction tools such as speech-to-text, text-to-speech, and object recognition, reflecting a typical pattern in architectures focused on perception and human interaction. In contrast, tools such as ontologies, neural networks, motivational systems, and

attention systems show co-occurrence values close to 0 with almost all others, indicating that they are used in isolation or in specific architectures.

1) SYMBOLIC AND DELIBERATIVE APPROACH

A significant number of architectures use PDDL as a language for symbolic knowledge representation and robot task planning. Specifically, this approach is present in papers FS2, FS3, FS4, FS5, FS8, FS9, FS10, FS11, FS13, FS16, FS17, FS20, and FS21. Several of these works also integrate planning systems such as ROSPlan (FS8, FS10, FS13, FS20) or PlanSys (FS11), which allow plans to be generated from the knowledge represented in PDDL. In some cases, such as FS10, FS20, and FS21, this symbolic knowledge is structured as knowledge graphs, which facilitates its integration with

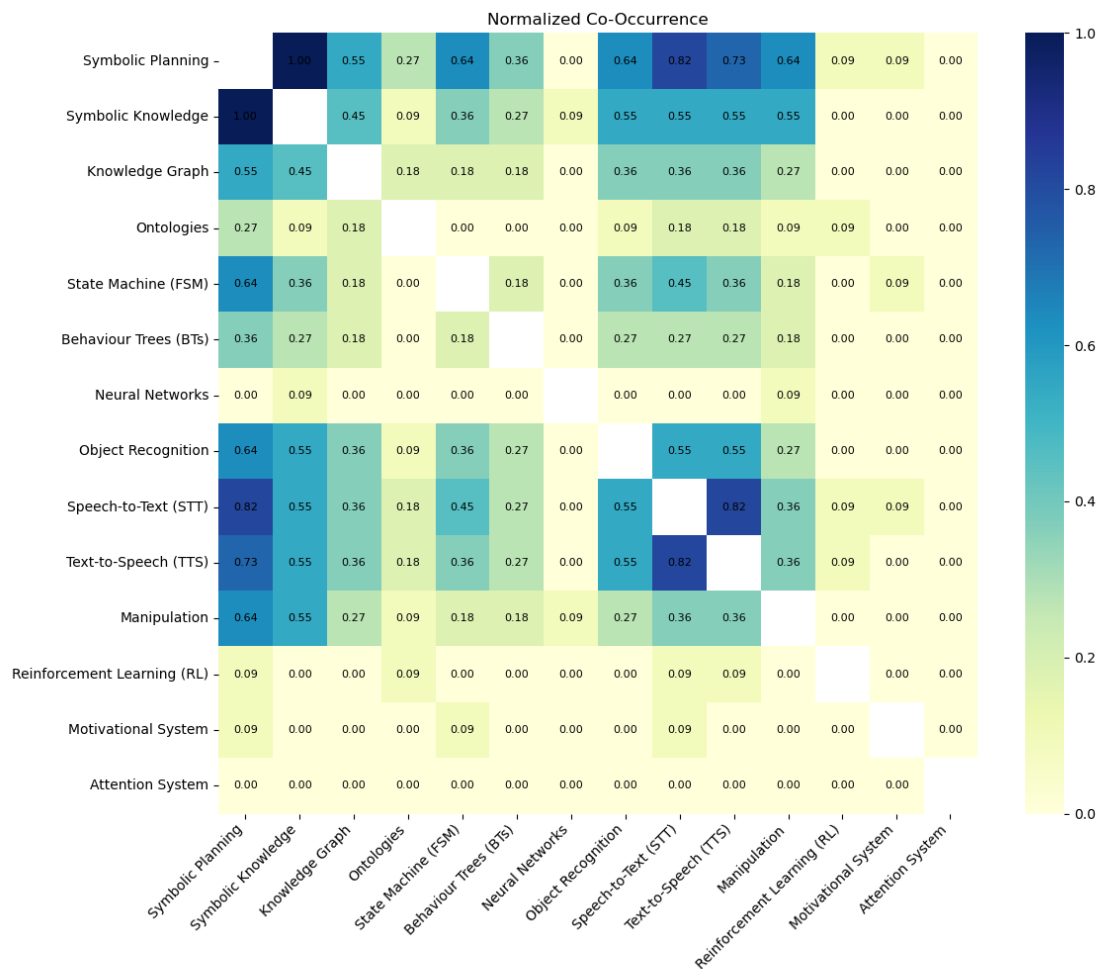


FIGURE 12. Heatmap co-occurrence diagram of the tools in the analyzed papers.

other modules. On the other hand, ontologies are used as support for symbolic reasoning, as seen in papers FS12 and FS13, and allow entities, relationships, and properties within the robot environment to be formally defined.

2) REACTIVE APPROACH WITH STATE MACHINES AND BEHAVIOR TREES

State machines are one of the most widely used mechanisms for executing both deliberative and reactive behaviors. They appear in the FS1, FS2, FS3, FS4, FS5, FS8, FS10, FS16, and FS18 architectures. In many of these cases, which are FS2, FS3, FS4, FS8, FS10, and FS16, state machines act as controllers of the plans generated by deliberative systems, but they also allow for the coordination of the entire architecture's operation. On the other hand, behavior trees are used in FS5, FS11, and FS21, generally to define the robot's actions based on the plans. In the case of FS11, for example, they are directly integrated with a deliberative system based on PDDL.

3) EMERGING APPROACH WITH NEURAL NETWORKS AND REINFORCEMENT LEARNING

Some architectures adopt emerging tools for specific tasks. The use of neural networks stands out in papers FS6, FS13,

and FS14. FS14, in particular, uses convolutional neural networks (CNN) and LSTM networks for human action recognition and behavior prediction. In FS6, a neural network controls the activation of previously symbolically represented actions, while in FS13, symbolic elements are combined with perception based on neural networks. FS12 deserves special mention, as it uses Hierarchical Reinforcement Learning (HRL) to articulate communication between the deliberative system and the reactive system. Finally, FS18 uses LLMs to generate responses in natural language.

4) HYBRID ARCHITECTURES AND CONVERGENCE OF APPROACHES

The combination of symbolic, reactive, and emergent tools gives rise to hybrid architectures such as FS5, FS6, FS10, FS12, FS13, FS20, and FS21. These architectures integrate various types of behavior representation and execution. For example, FS5 combines a symbolic graph with planners, state machines, and behavior trees, while FS6 symbolically represents actions but executes them using a neural network. In the case of FS4, a motivational system is introduced that modulates both deliberative goals and reactive behavior.

TABLE 9. Data extracted from papers accepted for the question on architecture evaluation. For each paper, its identifier is shown along with a paragraph summarizing how the evaluation was carried out.

ID	Architecture Evaluation
FS1	To evaluate the architecture, an experiment was designed in a simulated environment in which several participants had to teleoperate the robot from a starting point to an end point. In this experiment, there are two groups of participants: those with high cognitive abilities and those with low cognitive abilities.
FS2	Three types of experiments were conducted to evaluate the architecture. The first is a quantitative evaluation of the object recognition system. The second is based on testing the object learning feature. The third is a qualitative analysis of the entire system in which a user interacts with the robot by attempting to teach it the categories of different objects.
FS3	The experiment is based on a question game. The robot has a series of questions. When the experiment begins, the robot and a person start playing until a bell rings. Then, the robot has to change the mission.
FS4	Three experiments have been conducted with three different robots. The first uses a robotic arm to pick up and place objects. The second uses a simulated rover to acquire images. The third uses a surveillance robot also to acquire images, but this time taking into account the battery.
FS5	Three experiments have been conducted in this paper. The first experiment is based on generating behaviors for a robot that will make announcements. The robot will approach people and invite them to visit a panel. The second experiment includes assessing the degree of autonomy of elderly people. The third experiment is carried out in an apartment where the robot has to attend to the needs of a person with limited autonomy, such as bringing objects to that person.
FS6	Three experiments have been carried out. The first tests the ability to avoid obstacles, the second focuses on the execution of hierarchical tasks based on making deliveries, and the third is based on the concurrent execution of multiple plans.
FS7	The evaluation consists of using a group of participants who can interact with the robot.
FS8	An evaluation was carried out based on controlling a robot that has to patrol a series of points in a house. To do this, a simulated environment was used in which a robot is controlled with the architecture. The test was repeated with three different numbers of points to patrol.
FS9	The evaluation of this paper is based on tests involving a human participant. It begins with the human asking the robot to place a utensil on the table and ends when the robot completes the task.
FS10	A service robot is used in a home to validate the architecture. The robot has to notify one of the people in the house that they will receive a package.
FS11	Two experiments have been carried out. The first evaluated the robot's social navigation. To do this, the robot approaches a person. Then the human changes their orientation and the robot has to approach that person again. In the second experiment, the robot approaches the person to talk to them, and the person asks it to navigate to certain points or to come closer.
FS12	Validation is based on using a group of participants who interact with the robot through dialogue. The task consists of finding a word using sets of easy and difficult words.
FS13	The validation presented in this paper has been divided into two parts. The first experiment deals with the moment when the alarm goes off that a person needs assistance. The second experiment evaluates task prioritization in a scenario where the robot is asked to perform multiple tasks, such as closing a gas valve, bringing a book, opening the curtains, and monitoring the house.
FS14	The evaluation carried out in this paper has two phases. In the first phase, the performance of the different models is evaluated separately, i.e., the CNN and the LSTM. In the second phase, two different experiments are carried out. In the first experiment, the robot and a user have to assemble a wooden box. In the second experiment, the user and the robot have to assemble a small wooden chair. This experiment was conducted with three different users.
FS15	The evaluation of this paper is based on real-world testing with a Baxter humanoid robot. Experiments involved human-robot interaction in a lab setting, where the robot performed tea-making and sandwich-making tasks based on verbal cues. Multiple scenarios were tested with different object positions on the table.
FS16	The experiment consisted of two phases: first, simulation tests using ROS with seven user profiles and five robot profiles to evaluate the adaptive affective system in 35 scenarios; second, physical interactions with a real robot in a controlled area, using three user profiles and three robot profiles.
FS17	The evaluation has been divided into two phases. The first phase consists of patrolling a series of points in a house in a simulated environment. The second phase consists of performing the RoboCup@Home Carry My Luggage test in a real environment, interacting with a person.
FS18	Proof-of-concept scenarios have been carried out by running five real scenarios using two robotic platforms. These experiments involved interaction with humans through speech recognition, navigation, and manipulation. A study was also conducted with 30 participants to compare human interpretation of correctional language with robot behavior.
FS19	A study was conducted with 60 participants to evaluate external users' perceptions of the explainability and transparency system. Participants watched videos of the robot's interactions in three contexts (shopping mall, restaurant, and home), highlighting the robot's visual, verbal, and body language explanations when making decisions in conversation. Opinions were collected using Likert scale questionnaires on attributes such as warmth, competence, trust, and satisfaction with explanations. The data were statistically analyzed using mixed models, correlations, and mediation analysis.
FS20	The evaluation was carried out in real and simulated environments. Using a real robot, the experiment involved having the robot pick up objects and move them to the storage area. On the other hand, the simulation experiment involved solving the Tower of Hanoi puzzle and preparing a meal in a simulated kitchen.
FS21	The experiment was based on deploying the robot in a real-life scenario in which it had to act as a bartender. There were 116 participants who, standing in front of the robot, spoke directly to it to place orders.
FS22	Functional tests based on use cases were carried out. Two real-life assistance scenarios were implemented in a smart home and a nursing home. The first test focused on fall detection and response and coordination of robotic intervention. The second test focused on transporting objects by assisting a user with a walker to transport small objects using a robot.

FS16, on the other hand, integrates a motivational system based on emotional states and homeostatic decisions, demonstrating a more biologically inspired approach.

5) COMMON FUNCTIONAL COMPONENTS

In addition to planning and control mechanisms, many architectures include specialized modules for localization, navigation, and teleoperation, as seen in FS1, FS5, FS8, and FS16. Object recognition appears as a cross-cutting tool in FS2, FS5, FS8, FS16, FS20, and FS21, and is key to environmental perception. Similarly, speech recognition and speech synthesis are integrated into FS5, FS8, FS9, FS12, FS16, FS18, FS20, and FS21, providing human-robot

interaction capabilities. Finally, some architectures (FS9, FS16, and FS21) incorporate tools such as MoveIt to control the robot's manipulators, which is essential in tasks that require physical manipulation of the environment.

C. RQ3-EVALUATION METHODS FOR COGNITIVE ARCHITECTURES

After extracting and analyzing the data from the 22 selected papers, whose analysis is presented in Table 9, different types of cognitive architecture evaluation were found, which can be classified into two main groups. Thus, the following types of evaluation were found, which are presented in Figure 13. The basic types of evaluation found vary depending on

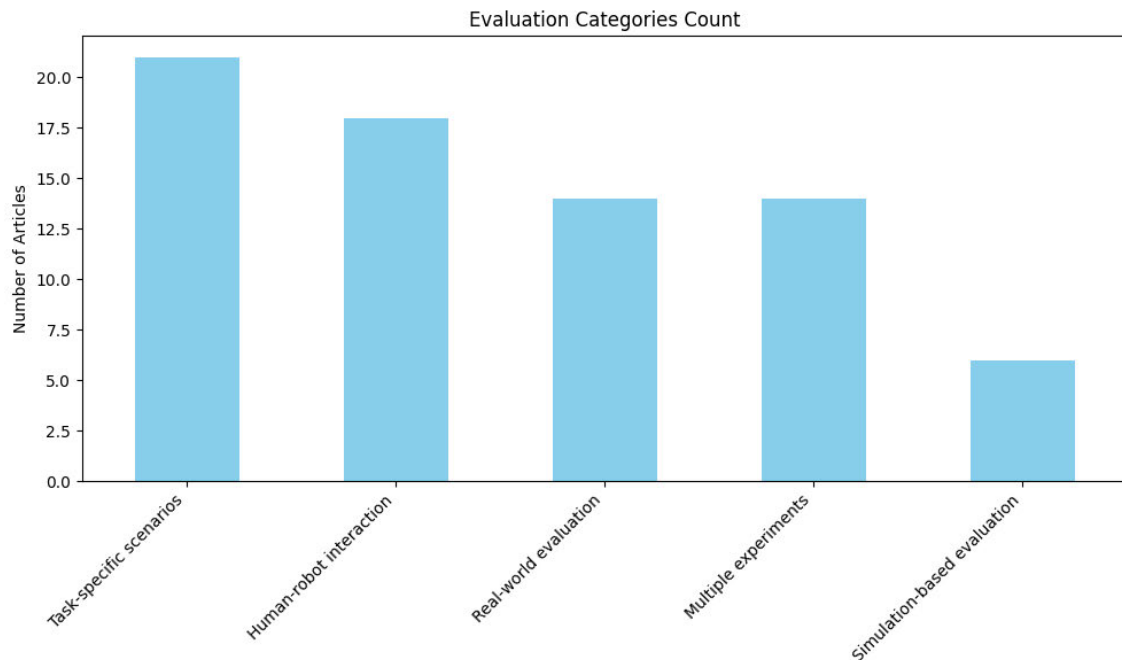


FIGURE 13. Frequency of occurrence of evaluation types.

whether they involve interaction with humans or scenarios with other tasks. On the other hand, a real robot or a simulated environment was used. Finally, we also took into account whether the evaluation involved repetition of experiments.

The co-occurrence diagram in Figure 14 shows how the different evaluation categories used in the analyzed papers relate to each other. The values represent the relative frequency of co-occurrence between pairs of categories, normalized between 0 and 1. This normalization was performed by dividing all values by the maximum co-occurrence value observed. It can be seen that the Task-specific scenarios category has a very high co-occurrence with Human-robot interaction and Real-world evaluation, indicating that these categories tend to appear together frequently. Similarly, Multiple Experiments also has a high co-occurrence with Task-specific scenarios and Real-world evaluation, suggesting that more comprehensive studies often involve real-world environments with multiple tasks. In contrast, Simulation-based evaluation has lower co-occurrence with other categories, especially Human-robot interaction, suggesting that simulated environments rarely include meaningful human interaction. This pattern indicates a tendency in the literature to separate technical evaluations in simulation from those focused on interaction or applied in real-world environments.

1) EVALUATIONS WITH HUMAN INTERACTION

A significant portion of the papers involves evaluations with human users, either through direct interaction or passive participation. For example, some studies focus on physical collaboration between humans and robots, such as in FS14,

where the robot and a user assemble a box and a wooden chair, or in FS13, where the prioritization of tasks requested by a person in a smart home is evaluated, or in FS15 where a robot has to perform tea-making and sandwich-making following verbal cues from a human.

Other research studies conversational or assistance interaction. In FS3, the robot plays a question game with the human until the mission changes after a bell rings. In FS5, three experiments are presented: one where the robot announces events to people, another where the autonomy of older people is evaluated, and another where the needs of a person with limited mobility are addressed. In FS10, the robot must notify a resident of the arrival of a package.

There are also more structured evaluations with groups of participants. In FS1, users remotely operate the robot and their cognitive abilities are compared. In FS7 and FS12, how multiple participants interact with the robot is analyzed. For example, FS12 focuses on a dialogue task to find words, and FS19 analyzes how users perceive the explanations given by the robot through videos and questionnaires. Similarly, FS21 employs 116 participants in a real environment where the robot acts as a bartender and takes orders verbally.

The study of social navigation is also present. In FS11, the robot approaches and reorients itself toward humans based on their movements. In FS18, speech recognition and navigation with humans are tested in five real scenarios, and language interpretation is evaluated by 30 participants.

Finally, some studies combine simulated and real environments. For example, FS16 performs patrols in simulation and then in a real environment with humans. FS22 implements

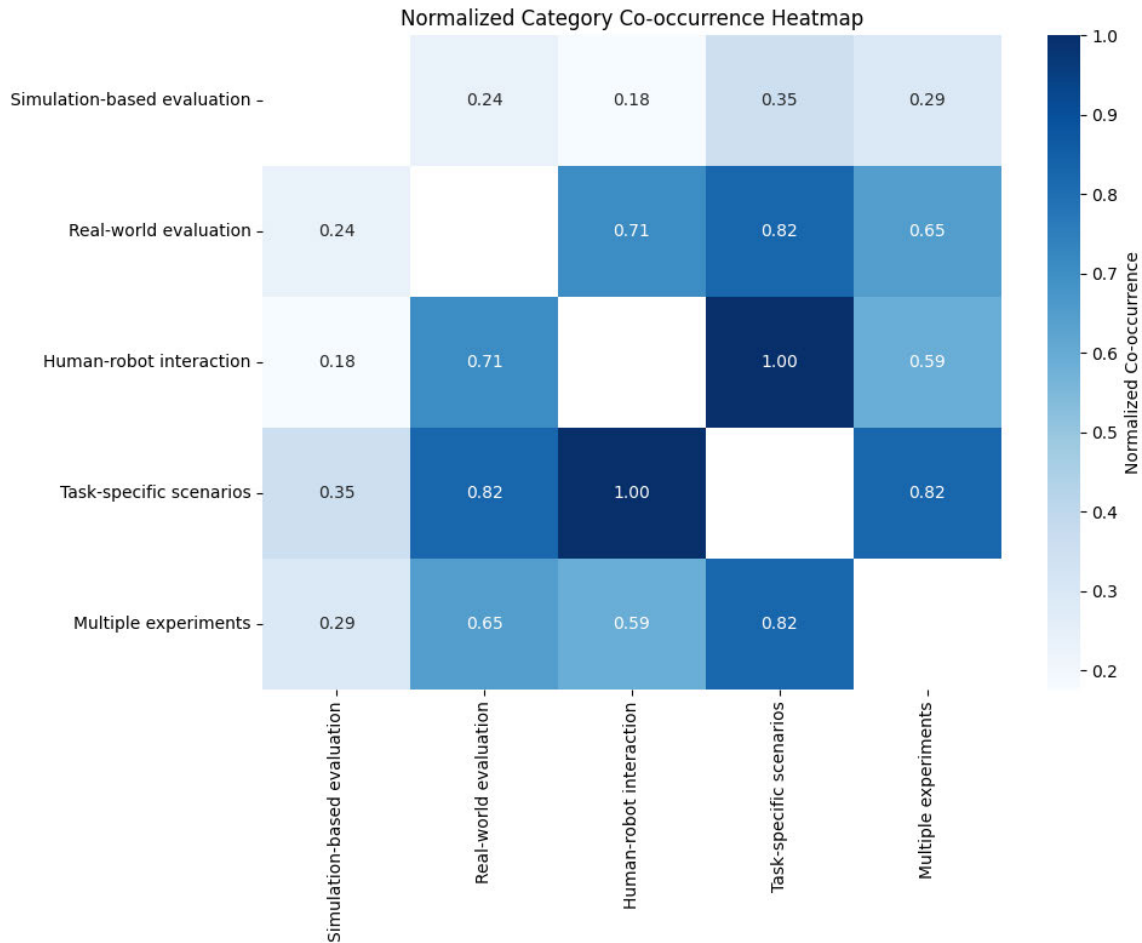


FIGURE 14. Heatmap co-occurrence diagram of the evaluation types of the analyzed papers.

two assistance scenarios: one for fall detection and another for assistance with objects using a robot.

2) EVALUATIONS WITHOUT HUMAN INTERACTION

In other cases, the evaluation is performed without direct human interaction, either through simulation or tests focused on specific components.

Some studies focus on individual components of the architecture. For example, FS2 performs quantitative evaluations of the object recognition system and tests for learning new categories. In FS6, obstacle avoidance, hierarchical task execution, and the execution of multiple concurrent plans are evaluated. FS14 also separates the validation of models, such as CNN and LSTM, before performing integrated tests with users.

Other research evaluates the entire architecture in simulated scenarios. In FS4, three platforms (robotic arm, rover, and surveillance robot) are used for tasks such as object manipulation and image acquisition, considering aspects such as battery life. In FS8 and FS17, a house is patrolled in a simulated environment. FS20 features tasks such as grasping real objects and, in simulation, solving the Towers of Hanoi

or preparing food. FS16 also includes testing with different user and robot profiles in simulation before moving on to real interaction.

D. RQ4-EVALUATION METRICS FOR COGNITIVE ARCHITECTURES

Evaluations of cognitive architectures use a wide variety of metrics. Table 10 presents the metrics used by each paper analyzed, both quantitative and qualitative. The graph in Figure 15 shows the frequency with which different categories of metrics were used in the papers analyzed. It can be seen that the most common metrics are those related to execution time and qualitative evaluations based on questionnaires. This suggests a strong concern for both the temporal performance of the systems and the perception or experience of the users. Other notable categories include interaction metrics, distance traveled, and accuracy/classification metrics, indicating significant interest in the robot's behavior during navigation and its learning ability. Emerging categories such as navigation safety, learning metrics, and emotional/behavioral assessments reflect growing attention to human-robot interaction quality and

TABLE 10. Data extracted from papers accepted for the question on metrics used in the evaluation. For each paper, its identifier and a paragraph summarizing the metrics used in the evaluation are shown.

ID	Metrics
FS1	To evaluate the experiments, qualitative measurements were taken using questionnaires completed by participants, and quantitative measurements were taken, such as the total time to complete the experiments and the time between collisions.
FS2	The metrics used are the number of categories learned, the number of questions and interactions to learn the categories, the average number of instances stored per category, and the overall classification accuracy.
FS3	The metrics used are based on internal motivational thresholds (saturation/satiation levels) and the derived Excitatory Potential (EP) value, which triggers behavior changes. Evaluation is conducted by tracking the activation of behaviors in response to simulated internal and external stimuli under different robot roles, with a focus on functional validation rather than performance benchmarks or user studies.
FS4	The metrics in this paper are the time spent in the deliberative system, the time needed to complete the missions, and the distance traveled.
FS5	Qualitative metrics are based on questionnaires completed by participants. The quantitative metric is the number of successful executions.
FS6	The metrics used are execution time, accuracy in avoiding obstacles, deliveries made, and erroneous plans.
FS7	Qualitative metrics were used through questionnaires completed by participants.
FS8	The metrics used were the execution time and the distance traveled by the robot during the evaluation tests.
FS9	Different concepts were measured during the experiments with this architecture. These are robustness, how many attempts the robot needs to perform the task, the time spent on reasoning, and the explainability of the architecture, whether it is possible to explain the robot's decisions.
FS10	The metrics used are mission execution time, distance traveled during the mission, calls to the planner, and the number of actions executed.
FS11	The metrics used are the minimum distance to the human during navigation, the distance traveled, the navigation time, and intrusions into the person's personal space. In addition, CPU usage, network usage, and memory usage were also used.
FS12	Qualitative metrics based on questionnaires were used to evaluate usability, interaction duration, the number of turns taken during the exercises, and the number of errors in the dialogue.
FS13	The metric used is the time it takes to execute the plans.
FS14	In the case of separate model evaluation, metrics such as confusion matrices, root mean square error (RMSE), and mean absolute error (MAE) were used. In the case of user experiments, the metrics used were the average number of tasks completed and the average time the user waited.
FS15	The main metric referenced is the Jaccard Similarity Index, used to calculate the semantic similarity between spoken words and available objects, which serves as the decision criterion for task selection. Validation is demonstrated through multiple experimental setups with varied object positions, showing consistent task execution aligned with the intended context rather than through numerical performance benchmarks.
FS16	The metrics used in the experiment included mean valence, which measures the robot's average emotional positivity during interaction; standard deviation of valence, which reflects emotional stability and responsiveness; and the percentage of activation of internal motivations (playing, resting, socializing), which indicates how internal needs guide behavior. The percentage of activation of emotional states was also measured to assess affective expressiveness, the evolution of valence over time to observe reactions to different user profiles, and, in the adaptive settings, the evolution of the weights between the internal state and the interaction, to assess the robot's adaptability.
FS17	For the first phase of the simulator-based evaluation, the metrics used were execution time and distance traveled. For the second phase, based on a real environment, the metrics used were the time to generate PDDL, the time spent planning, and the time between the architecture receiving the objectives and generating the plan. Finally, performance metrics such as CPU load, RAM, MB Sent, and Packages Received were also included.
FS18	The metrics used in this evaluation are the rate of agreement with human responses, the rate of agreement on specific tasks, and the success of the robot's missions.
FS19	The metrics used are based on the explainability study carried out. This provides an F1 score for evaluating the components of the architecture. On the other hand, questionnaires were used to evaluate the explainability of the architecture.
FS20	The metrics used in this work are the average depth of the network, planning time, execution time, movement planning time, and the number of movement planning attempts.
FS21	The metrics are beverage preparation times, response times, voice and intention recognition percentage, and dialogue duration. There are also user experience questionnaires (UEQ) and the Unified Theory of Acceptance and Use of Technology (UTAUT).
FS22	The metrics for this work are object detection update frequency, knowledge graph synchronization time, overall latency (knowledge graph agent update propagation), GPU usage, CPU usage, and power consumption.

adaptive capabilities. On the other hand, metrics such as explainability, deliberative aspects, and emotional/behavioral evaluations appear less frequently, which could point to areas that are still under-explored or of lower priority in current evaluations.

Going into more detail in the papers, the execution time of missions or tasks (FS1, FS4, FS6, FS8, FS10, FS11, FS13, FS17, FS20, FS21) is a very common metric. The distance traveled by the robot (FS4, FS8, FS10, FS11, FS17) and deliberative metrics, including time spent in reasoning systems, planning time, and calls to the planner (FS4, FS9, FS10, FS17, FS20), are used to a lesser extent. Success rate and attempt-based metrics, such as the number of successful executions and task completion rates (FS5, FS6, FS10, FS14, FS18, FS20), are also common. System performance metrics, which now encompass resource usage such as CPU, RAM, network, and power consumption (FS11, FS17, FS22), provide insights into computational efficiency.

The analysis reveals significant trade-offs in current evaluation approaches that merit deeper consideration. The focus on execution time metrics, present in 10 studies,

often conflicts with mission success rates, creating tensions between efficiency and effectiveness. For instance, faster planning may lead to suboptimal plans, while comprehensive deliberation may exceed real-time constraints. This planning time versus mission success trade-off is particularly evident in hybrid architectures that must balance reactive responsiveness with deliberative thoroughness. Similarly, the quality of human-robot interaction presents complex trade-offs with task performance efficiency, as more natural and engaging interactions often require additional processing time and may reduce task completion rates.

Notable gaps exist in evaluation methodologies, particularly regarding explainability metrics, which appear in only 2 studies despite their critical importance for deployment in real-world applications. Explainability metrics, when present (FS9, FS19), focus primarily on decision traceability rather than comprehensibility to end users or domain experts. These underexplored areas represent significant opportunities for advancing the evaluation methodologies and ensuring trustworthy cognitive systems.

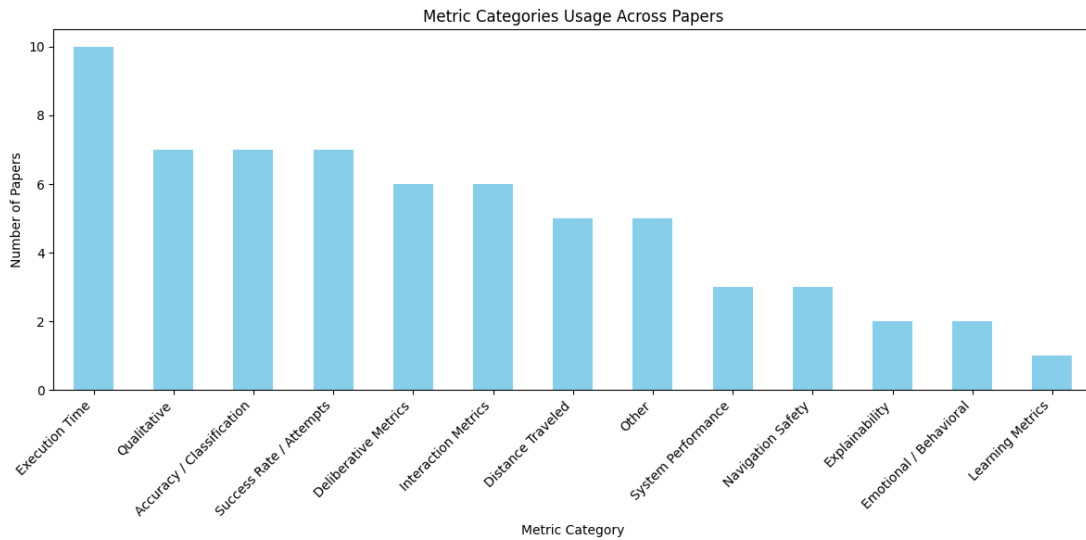


FIGURE 15. Summary of the most commonly used metrics in the cognitive architecture evaluations of the analyzed papers.

Some evaluations consider interaction metrics, which now encompass both user interaction and communication aspects. For example, papers FS2, FS11, FS12, FS14, FS16, and FS21 use various interaction measures, including dialogue duration, turn-taking, voice recognition, spatial interaction patterns, and user experience metrics such as waiting times and subjective time perception. Additionally, qualitative evaluations through questionnaires are employed by papers FS1, FS5, FS7, FS12, FS14, FS19, and FS21 to measure aspects such as usability, explainability, user experience, and technology acceptance.

Deliberative metrics encompass various planning and reasoning aspects, including calls to the planner, planning time, PDDL generation time, reasoning time, and planning quality assessments such as erroneous plans (FS4, FS6, FS9, FS10, FS17, FS20). Learning metrics are employed by papers FS2 to evaluate capabilities such as the number of categories learned, questions/interactions required for learning, and classification accuracy. Navigation safety metrics, including collision avoidance and personal space considerations, are used in papers FS1, FS6, and FS11. Additionally, emotional and behavioral metrics are employed by papers FS3 and FS16 to assess internal motivational states and affective expressiveness.

However, several papers employ specialized metrics that do not fit into the previous categories, representing the Other category in our classification. Below are these specialized metrics:

- Functional validation approaches that focus on behavior activation tracking rather than performance benchmarks (FS3).
- Robustness assessments measuring system resilience and reliability under varying conditions (FS9).
- Context-based validation approaches using semantic similarity for task selection rather than traditional performance metrics (FS15).

- Architectural metrics such as network depth and structural complexity measures (FS20).
- System synchronization metrics, including update frequencies, knowledge graph synchronization times, and propagation latencies (FS22).

Together, these metrics make it possible to evaluate everything from technical performance to the quality of human-robot interaction and the adaptability and learning capacity of cognitive architectures. This is essential for characterizing not only computational efficiency or task execution capacity, but also more complex and human aspects, such as collaboration, interpretation of intentions, explanation of decisions, and the emotional impact of systems. Furthermore, by considering both simulated environments and experiments with real users, a comprehensive view of the architecture's capabilities in different contexts is achieved, contributing to its continuous improvement and potential deployment in real-world applications.

VI. DISCUSSION

This systematic review presents a comprehensive analysis of cognitive architectures in autonomous robotics, focusing on behavior generation mechanisms. The analysis of 22 high-quality studies spanning 2018-2025 reveals significant patterns, emerging challenges, and opportunities that shape our understanding of this evolving field. The findings illuminate both current accomplishments and critical gaps that require attention from researchers and practitioners. Table 11 provides a comprehensive synthesis of the analyzed cognitive architectures, revealing clear patterns in technological infrastructure, architectural choices, and evaluation methodologies.

A. ARCHITECTURAL EVOLUTION AND HYBRID PARADIGMS

The prevalence of hybrid architectures (59.1% of reviewed studies) presents a significant evolution in cognitive system

TABLE 11. Overview of Cognitive Architectures: Architecture Type, middleware, evaluation methods, and metrics.

ID	Architecture	Middleware	Evaluation	Metrics
FS1	reactive/emergent	ROS	Simulation, HRI	Qualitative, Execution Time, Navigation Safety
FS2	hybrid	ROS	HRI, Task-specific, Multiple exp.	Accuracy/Classification, Interaction Metrics, Learning Metrics
FS3	hybrid	ROS	HRI, Task-specific	Emotional/Behavioral, Other
FS4	hybrid	ROS	Simulation, Task-specific, Multiple exp.	Execution Time, Distance Traveled, Deliberative Metrics
FS5	hybrid	RoboComp	Real-world, HRI, Task-specific, Multiple exp.	Qualitative, Success Rate
FS6	hybrid	ROS	Task-specific, Multiple exp.	Execution Time, Accuracy/Classification, Success Rate, Deliberative Metrics, Navigation Safety
FS7	reactive/emergent	ROS	HRI	Qualitative
FS8	hybrid	ROS	Simulation, Task-specific, Multiple exp.	Execution Time, Distance Traveled
FS9	hybrid	ROS	Real-world, HRI, Task-specific	Success Rate, Deliberative Metrics, Explainability, Other
FS10	hybrid	ROS	Real-world, HRI, Task-specific	Execution Time, Distance Traveled, Deliberative Metrics, Success Rate
FS11	hybrid	ROS 2	Real-world, HRI, Task-specific, Multiple exp.	Execution Time, Distance Traveled, System Performance, Interaction Metrics, Navigation Safety
FS12	hybrid	ROS	HRI, Task-specific	Qualitative, Interaction Metrics
FS13	deliberative/symbolic	ROS	Real-world, HRI, Task-specific, Multiple exp.	Execution Time
FS14	reactive/emergent	ROS	Real-world, HRI, Task-specific, Multiple exp.	Qualitative, Accuracy/Classification, Success Rate, Interaction Metrics
FS15	deliberative/symbolic	ROS	Real-world, HRI, Multiple exp.	Accuracy/Classification, Other
FS16	reactive/emergent	ROS	Simulation, Real-world, HRI, Task-specific, Multiple exp.	Interaction Metrics, Emotional/Behavioral
FS17	hybrid	ROS 2	Simulation, Real-world, HRI, Task-specific, Multiple exp.	Execution Time, Distance Traveled, System Performance, Deliberative Metrics
FS18	deliberative/symbolic	ROS	Real-world, HRI, Task-specific, Multiple exp.	Accuracy/Classification, Success Rate
FS19	reactive/emergent	YARP	HRI, Task-specific	Qualitative, Accuracy/Classification, Explainability
FS20	deliberative/symbolic	ROS	Simulation, Real-world, Task-specific, Multiple exp.	Execution Time, Success Rate, Deliberative Metrics, Other
FS21	hybrid	ROS	Real-world, HRI, Task-specific	Qualitative, Execution Time, Accuracy/Classification, Interaction Metrics
FS22	hybrid	ROS 2/RoboComp	Real-world, HRI, Task-specific, Multiple exp.	System Performance, Other

design approaches within the robotics community. This trend reflects the understanding that real-world autonomous robotics requires solutions that combine symbolic and subsymbolic approaches.

As presented in Table 11, not only do hybrid architectures dominate numerically but also exhibit the most sophisticated evaluation methodologies, with most hybrid systems (FS2, FS4, FS5, FS6, FS9, FS10, FS11, FS17, FS21, FS22) incorporating multiple experimental conditions and diverse metric categories. This pattern suggests that hybrid architectures have matured to the point where comprehensive evaluation is both feasible and necessary, contrasting with reactive/emergent architectures that often rely on more limited evaluation approaches (FS1, FS7, FS16, FS19) and deliberative/symbolic architectures that tend to focus on specific performance metrics (FS13, FS15, FS18, FS20).

The shift toward hybrid architectures stems from the recognition that pure deliberative architectures, while effective for logical reasoning and long-term planning, face challenges in dynamic environments requiring real-time responsiveness. The computational overhead of symbolic reasoning, combined with the symbol grounding problem [8], creates limitations in unstructured, time-sensitive scenarios. Conversely, reactive architectures, though capable of rapid environmental response, lack the reasoning capabilities necessary for complex goal hierarchies and multi-step planning tasks.

Hybrid architectures provide a solution that preserves the computational advantages of reactive systems while maintaining access to the symbolic reasoning capabilities.

The temporal analysis shows increasing hybrid adoption (2023-2025), indicating that this represents a convergence toward architectures that can handle the spectrum of cognitive robotics challenges. This evolution parallels developments in cognitive science, where hybrid models of human cognition have gained acceptance for their ability to explain both fast, intuitive responses and slower, deliberative reasoning processes.

Moreover, hybrid architectures require coordination mechanisms between symbolic and subsymbolic components, creating challenges in knowledge representation, learning transfer, and system integration. The implementations identified in this review demonstrate that these challenges can be addressed, but they also reveal the need for principled approaches to hybrid system design.

B. TECHNOLOGICAL INFRASTRUCTURE AND STANDARDIZATION PATTERNS

The adoption of ROS/ROS 2 (90.9% of studies) represents an achievement in technological standardization within the robotics community. This convergence creates both opportunities and challenges that extend beyond tool selection. The standardization facilitates improved reproducibility, code sharing, and collaborative development, addressing challenges in robotics research where custom middleware often hinders knowledge transfer between research groups.

The middleware distribution patterns revealed in Table 11 demonstrate remarkable convergence, with ROS appearing in 17 studies and ROS 2 in 3 studies (including one hybrid system), while alternative middleware (RoboComp,

YARP) appears in only 2 studies total. This concentration is particularly notable when considering that middleware choice often influences architectural design decisions and available integration tools. The transition from ROS to ROS 2, derived from the EOL of ROS 1 [75], demonstrates the community's ability to evolve while maintaining compatibility, but it also highlights the challenges of technological migration in a field with infrastructure investments. Furthermore, the table shows that the few studies employing alternative middleware (FS5 with RoboComp, FS19 with YARP, FS22 with ROS 2/RoboComp hybrid) do not exhibit systematically different evaluation approaches or architectural characteristics. This suggests that the core cognitive architectural principles are largely middleware-independent, though the implementation strategies may vary.

The tool co-occurrence analysis reveals an ecosystem that has evolved around this middleware foundation. The emergence of a canonical architectural pattern, combining PDDL-based symbolic planning, state machines, or behavior trees for execution control, and standardized perception and speech interfaces, represents a mature engineering approach to cognitive system design. This pattern provides researchers with a proven foundation while potentially constraining exploration of different architectural approaches.

The isolation of advanced techniques such as ontologies, neural networks, and motivational systems in the co-occurrence analysis suggests that the field of cognitive architectures has not fully integrated these tools into mainstream cognitive architectures. This represents both a limitation of current practice and an opportunity for future innovation.

C. EVALUATION METHODOLOGY CHALLENGES AND STANDARDIZATION OPPORTUNITIES

The heterogeneity in evaluation approaches identified in this review represents a challenge facing the cognitive robotics community. While this diversity reflects the nature of cognitive robotic systems and their varied application domains, it undermines the ability to build cumulative knowledge and make comparisons between different approaches.

Table 11 reveals systematic patterns in evaluation methodology adoption across architectural types. Hybrid architectures demonstrate comprehensive evaluation strategies, with 8 of 13 studies incorporating multiple experimental conditions and most combining simulation, real-world testing, and human-robot interaction. This comprehensive approach contrasts with reactive/emergent architectures, where only 2 of 5 studies employ multiple experimental conditions, and deliberative/symbolic architectures, which show high comprehensiveness, with all 4 studies using multiple conditions but often with a narrower scope. This pattern suggests that architectural complexity may drive evaluation comprehensiveness, but it also indicates potential bias in how different architectural approaches are validated and compared in the literature.

The absence of standardized benchmarks contrasts with other AI domains, where common evaluation frameworks have accelerated progress. In computer vision, datasets have enabled systematic comparison of algorithms and tracking of field-wide progress. Natural language processing has benefited from shared tasks and standardized evaluation metrics. Cognitive robotics lacks such unifying frameworks, largely due to the complexity and cost of robotic systems, but also because of community coordination around evaluation standards.

The tool distribution analysis in Table 11 reveals concerning patterns in how cognitive capabilities are evaluated. Execution time appears as a metric in 10 studies (45.5%), while success rate appears in 7 studies (31.8%), and explainability metrics in merely 2 studies (9.1%). This distribution is particularly problematic given that 18 of 22 studies (81.8%) involve human-robot interaction or real-world deployment scenarios where explainability, robustness, and safety would be critical factors. The table shows that system performance metrics (CPU, memory, network usage) appear in only 3 studies (13.6%), despite the computational complexity inherent in cognitive architectures. Most concerning is the near-absence of metrics specifically designed to evaluate cognitive functions such as learning transfer, knowledge retention, or adaptive reasoning, suggesting that the field may be evaluating cognitive architectures using criteria more appropriate for conventional control systems.

The limited focus on long-term evaluation represents a significant concern, as cognitive architectures are fundamentally designed for continuous learning and adaptation. Most studies evaluate performance over short time horizons. This limitation stems from both practical constraints in conducting longitudinal studies and current gaps in methodologies for evaluating long-term cognitive development in artificial systems.

D. IMPLICATIONS FOR RESEARCH AND PRACTICE

The findings reveal that cognitive robotics has reached a pivotal point where the research community must balance theoretical advances with practical deployment requirements. The dominance of hybrid architectures (59.1% of studies) indicates that the community has moved beyond debates between symbolic and subsymbolic approaches toward pragmatic integration solutions.

For researchers, this presents clear opportunities and challenges. The main opportunity lies in developing better methods for combining different cognitive components while maintaining system performance and explainability. The primary challenge is addressing evaluation methodology gaps that limit systematic progress. Key research priorities include developing standardized evaluation frameworks for meaningful study comparisons, improving real-time symbolic reasoning capabilities, integrating modern machine learning with cognitive architectures, and establishing foundations for explainable cognitive systems.

The widespread adoption of ROS and ROS 2 (90.9% of studies) and the canonical tool patterns identified in this review create significant implications for technological infrastructure. This standardization accelerates development and provides practitioners with proven technological foundations. The common adoption of PDDL for planning, ROS for communication, and established perception and speech tools suggests that effective cognitive architectures can be built using proven components rather than requiring fundamental infrastructure innovation.

A critical implication concerns the need for evaluation methodology reform. The heterogeneous evaluation approaches identified represent a barrier to scientific progress that requires coordinated community action. Cognitive robotics research needs standardized benchmarks, evaluation protocols, and performance metrics that address not only technical performance but also safety, explainability, and human factors essential for real-world deployment.

The evolution toward more sophisticated cognitive capabilities raises important questions about societal implications. The limited attention to ethical considerations represents a gap that must be addressed as these systems become more capable. The development of socially-aware architectures with theory of mind, emotional modeling, and natural language interaction requires frameworks ensuring that cognitive systems respect human autonomy and dignity while maximizing beneficial potential.

VII. CONCLUSION

This systematic review presents a comprehensive analysis of cognitive architectures in autonomous robotics, focusing on behavior generation approaches and evaluation methodologies. By analyzing 22 high-quality publications from 2018 to mid-2025, this study maps the current landscape and identifies key trends, challenges, and opportunities for the field.

The analysis reveals some key insights about cognitive architectures in autonomous robotics. Hybrid architectures have emerged as the dominant paradigm, representing the field's recognition that complex real-world robotic tasks require integration of both symbolic reasoning and reactive responsiveness. The Robot Operating System (ROS and ROS 2) has achieved widespread adoption, establishing a de facto standard that facilitates reproducibility and knowledge transfer across research groups and industrial applications. A canonical design pattern has emerged involving symbolic planning (PDDL-based), state machines or behavior trees for execution control, and perception and speech modules that work in coordination to achieve autonomous behavior. While evaluation methodologies show diversity with execution time and user questionnaires being most common, systematic gaps exist, particularly in explainability metrics, indicating areas where the field must evolve to meet real-world deployment demands and user trust requirements.

This review makes several key contributions to the field by providing a comprehensive mapping focused specifically

on behavior generation in current cognitive architectures for autonomous robotics. The work documents the evolution toward hybrid architectures and identifies clear patterns in tool selection and integration that have emerged across different research groups. Additionally, it presents a systematic analysis of evaluation methodologies in this domain, revealing critical gaps and opportunities for improvement that can guide future research toward more rigorous assessment protocols and standardized benchmarking approaches.

Based on the analysis of current trends and identified gaps, several critical research directions emerge as essential for advancing cognitive robotic systems. The development of standardized evaluation frameworks represents a fundamental need, requiring unified protocols that move beyond execution-time metrics toward comprehensive assessment of adaptability, continual learning, and explainability. This includes developing longitudinal evaluation protocols and standardized datasets that capture the multifaceted nature of cognitive capabilities in dynamic environments.

The integration of modern reasoning paradigms, particularly Large Language Models and neural-symbolic methods, offers potential for balancing natural language understanding, perception, and symbolic planning. Critical areas include developing hybrid reasoning systems and robust grounding mechanisms between linguistic representations and robotic actions, enabling more intuitive human-robot interaction and sophisticated decision-making processes.

Explainability and trust emerge as fundamental requirements rather than optional features, necessitating interpretability as a core design principle rather than a post-hoc addition. This involves developing multi-level explanation systems and natural language generation capabilities for complex reasoning chains, ensuring cognitive systems can articulate their decision-making processes to diverse stakeholders with varying technical backgrounds.

The challenge of continual and ethical learning requires addressing lifelong learning without catastrophic forgetting while incorporating social norm understanding and ethical alignment into system behavior. Research should focus on memory architectures that selectively preserve knowledge and cognitive systems that respect human autonomy and adapt appropriately to different cultural contexts and social situations.

Finally, real-time cognitive optimization demands innovative hardware and software strategies that enable cognition under strict timing constraints. This includes exploring neuromorphic computing implementations, edge-cloud hybrid processing models, and approximate reasoning algorithms that maintain cognitive capabilities while meeting the temporal requirements of real-time robotic operation.

The field of cognitive architectures for autonomous robotics is transitioning from a research discipline focused on individual components to an engineering discipline concerned with system integration and real-world deployment. The convergence around hybrid architectures (59.1% of studies) and standardized tools represents significant

TABLE 12. PRISMA 2020 checklist for this systematic review.

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract section
INTRODUCTION			
Rationale	3	Describe the rationale for the review.	Introduction
Objectives	4	Provide explicit statement of objectives.	Introduction, Section III-B
METHODS			
Eligibility criteria	5	Specify inclusion/exclusion criteria.	Section III-E, Tables 2-3
Information sources	6	Specify all databases searched.	Section III-C1, III-D
Search strategy	7	Present full search strategies.	Section III-D
Selection process	8	Specify study selection methods.	Section III-E, Figure 5
Data collection process	9	Specify data collection methods.	Section III-G
Data items	10a	List and define all outcomes.	Section III-G
Study risk of bias assessment	11	List other variables sought.	Section III-F, Table 4
Effect measures	12	Specify bias assessment methods.	Section III-F
Synthesis methods	13a	Specify effect measures used.	Section III-G, Section V
	13b	Describe eligibility processes.	Section III-E, IV-A
	13c	Describe data preparation methods.	Section III-G
	13d	Describe display methods.	Section V, Figures 7-15
Reporting bias assessment	14	Describe synthesis methods.	Section V
Certainty assessment	15	Describe sensitivity analyses.	Section III-F
RESULTS			
Study selection	16a	Describe the selection process.	Section IV-A, Figure 6
	16b	Cite excluded studies.	Section IV-A, IV-B
Study characteristics	17	Cite included studies.	Table 6, Section V
Risk of bias in studies	18	Present bias assessments.	Section III-F, Table 5
Results of individual studies	19	Present results per study.	Tables 7-10
Results of synthesis	20a	Summarize characteristics.	Section V
	20b	Present statistical syntheses.	Section V, Figures 7-15
	20c	Present heterogeneity investigations.	Section V-D
Reporting biases	21	Present sensitivity analyses.	Section III-F
Certainty of evidence	22	Present bias assessments.	Section III-F
DISCUSSION			
Discussion	23a	General interpretation.	Section VI
	23b	Limitations of evidence.	Section VI-D
	23c	Limitations of process.	Introduction, Section III-C2
	23d	Implications.	Section VI-D, VII-C
OTHER INFORMATION			
Registration and protocol	24a	Registration information.	Not reported
	24b	Protocol access.	Not reported
	24c	Amendments.	Not applicable
Support	25	Funding sources.	Acknowledgment
Competing interests	26	Competing interests.	Not reported
Availability of materials	27	Data/code availability.	Not reported

progress toward mature, deployable systems, yet important challenges remain in evaluation methodology, advanced learning integration, and explainability that require sustained research attention.

This systematic review provides a foundation for the research community to address these challenges and advance the development of more intelligent, robust, and trustworthy autonomous robotic systems. The insights and directions identified here are essential for realizing the full potential of cognitive robotics in addressing complex real-world

challenges across diverse application domains, from health-care and manufacturing to space exploration and domestic assistance.

APPENDIX A DATA AND CODE

The datasets generated and the code used for analysis are available under a public license in the GitHub repository at https://github.com/mgonzs13/cognitive_architecture_systematic_review_2025.

APPENDIX B

PRISMA 2020 CHECKLIST

This review was conducted in accordance with the PRISMA 2020 statement. The complete checklist is provided in Table 12.

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